Set 2 : Tests 1 to 10

1. Bhuta, Mumbai
2. Cresmonde World, Mumbai
3. Kohinoor International, Mumbai
4. Lilavatibai Podar, Mumbai
5. Orion, Mumbai
6. Pawar Public, Kandivli, Mumbai
7. Ramniwas Bajaj, Mumbai
8. St Francis, Mumbai
9. Thakur International, Mumbai
10. Jankidevi, Mumbai

Test 1

I – Prelim

Subject: Mathematics Class: X Duration: 3 hours Marks: 80

Name : _____ Roll No: _____

General Instructions:

1. ***This paper contains 7 number of pages.***
2. ***Answers to this paper must be written on the answer sheet provided.***
3. ***The first 15 minutes is to be spent in reading the question paper only.***
4. ***Rough Column must be drawn in the answer sheet. No marks will be given without rough column.***

SECTION – I (40 marks)

QUESTION: (1)

Choose the correct answers to the questions from the given options: [15]

(i) A recurring deposit account has a monthly installment of Rs 2500 for 1 ½ years. The money deposited is

(a) Rs 25000
(b) Rs 45000
(c) Rs 4500
(d) Rs 3000

(ii) Money to be invested to buy 400 shares of Rs 100 at 20% discount is

(a) Rs 32000
(b) Rs 40000
(c) Rs 80000
(d) Rs 8000

(iii) If $-5x \leq 150, x \in R$, the smallest value of x is

(a) 30
(b) 50
(c) -30
(d) -50

(iv) Assertion (A): A customer who is the end user files for GST refund.
Reason (R): The end-user does not have any claim on the GST paid by him/her.

(a) A is true, R is false.
(b) A is false, R is true.
(c) Both A and R are true.
(d) Both A and R are false.

(v) The roots of the quadratic equation $x^2 - 14x + 49 = 0$ are

(a) Real and distinct

(b) Distinct

(c) Imaginary

(d) Real and equal

(vi) If (x – 2) is a factor of $x^3 + 2x^2 - kx + 10$. The value of k is

(a) 13

(b) -13

(c) 12

(d) -12

(vii) The transpose of matrix A = $\begin{bmatrix} -2 & 3 \\ 0 & -4 \end{bmatrix}$ is

(a) $\begin{bmatrix} 2 & -3 \\ 0 & 4 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 0 \\ -3 & 4 \end{bmatrix}$

(c) $\begin{bmatrix} -2 & 0 \\ 3 & -4 \end{bmatrix}$

(d) $\begin{bmatrix} 8 \\ 0 \end{bmatrix}$

(viii) The 100th term of the sequence -20, -10, 0, 10, 20….. is

(a) 900

(b) 950

(c) 930

(d) 960

(ix) For the GP 3, 6, 12, 24, 48… the 10th term is

(a) 512

(b) 1536

(c) 1024

(d) 4608

(x) The mid-point of the line segment joining (4, -8) and (-6, -10) is

(a) (-1, -9)

(b) (5, 9)

(c) (-9, -1)

(d) (0, 0)

(xi) The centres of two circles are at a distance of 10 cm apart. If the radii of the circles are 6 cm and 2 cm, the length of the transverse common tangent is

(a) 10 cm

(b) 18 cm

(c) 4 cm

(d) 6 cm

(xii) The volume of a cone of r = 8 cm and h = 30 cm is

(a) 320π cm^3

(b) $640\,\pi$ cm^3

(c) $900\,\pi$ cm^3

(d) $240\,\pi$ cm^3

(xiii) $1 - sin^2\theta + 1 - cos^2\theta$ is equal to

(a) 0

(b) 1

(c) -1

(d) -2

(xiv) The mean of 20, 12, x, 8, 14 is 13.8. Then x is

(a) 15

(b) 18

(c) 20

(d) 13

(xv) When a die is thrown, the probability of getting a square number is

(a) 0

(b) 1/3

(c) 1/6

(d) 1/2

QUESTION: (2)

a. Salman deposits Rs 1200 every month in a recurring deposit account for 2½ years. If the rate of interest is 6% p.a, find the amount he will receive on maturity. [4]

b. If A(-4, 3) and B(8, -6), [4]

(i) Find the length of AB.

(ii) In what ratio is the line joining AB divided by the x-axis?

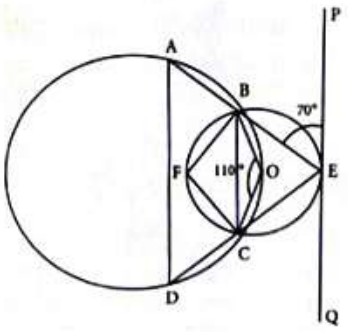
c. In the given diagram, ABCD is a cyclic quadrilateral and PQ is a tangent to the smaller circle at E. Given $\angle AEP = 70°$, $\angle BOC = 110°$. Find: [4]

(i) $\angle ECB$

(ii) $\angle BEC$

(iii) $\angle BFC$

(iv) $\angle DAB$

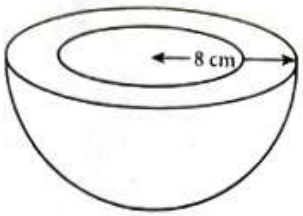


QUESTION: (3)

a. The inner circumference of the rim of a circular metal tub is 44 cm. Find: [4]

(i) The inner radius of the tub.

(ii) The volume of the material of the tub if its outer radius is 8 cm. Give your answer correct to three significant figures.



b. Find the value of a and b if x – 1 and x – 2 are factors of $x^3 - ax + b$. [4]

c. Use graph sheet for this question. Take 2 cm = 1 unit along the axes. [5]
Plot the ΔPAB, where O(0, 0), A(3, -2) and B(2, -3).

(i) Reflect the OAB through the origin and name it as OA'B'.

(ii) Reflect the OA'B' on the y-axis and name it as OA"B".

(iii) Reflect the OA'B' on the x-axis and name it as OA'''B'''.

(iv) Join the points A A" B" B' A' A''' B''' B and give the geometrical name of the closed figure so formed.

SECTION – II (40 marks)

QUESTION: (4)

a. The following bill shows the GST rates and the marked price of articles: [3]

Articles	Marked price	Rate of GST
Graphic card	Rs 15500	18%
Laptop adapter	Rs 1900	28%

Find the total amount to be paid for the above bill.

b. Solve the following quadratic equation, $7x^2 + 2x - 2 = 0$.
Give your answer correct to two places of decimal. [3]

c. Use a graph sheet for this question. Draw a histogram for the daily earnings of 54 medical stores in the following table and hence estimate the mode for the following distribution. Take 2 cm = Rs 500 units along the x-axis and 2 cm = 5 stores along the y-axis. [4]

Daily earnings (Rs)	4500-5000	5000-5500	5500-6000	6000-6500	6500-7000
No. of medical stores	20	14	12	5	3

QUESTION: (5)

a. Solve for x, using the properties of proportion. [3]

$$\frac{\sqrt{2+x}+\sqrt{3-x}}{\sqrt{2+x}-\sqrt{3-x}} = 3$$

b. Using ruler and compasses, construct a regular hexagon of 4.5 cm side. Hence construct a circle circumscribing the hexagon. Measure and write down the length of the circum-radius. [3]

c. The following table gives the petrol prices per litre for a period of 50 days. [4]

Price(Rs)	85-90	90-95	95-100	100-105	105-110
No. of days	12	10	8	15	5

Find the mean price of petrol per litre to the nearest rupee using the step-deviation method.

QUESTION: (6)

a. Given that: $M \times \begin{bmatrix} 3 & -4 \\ 0 & 2 \end{bmatrix} = [3 \quad 8]$. Find: [3]

(i) The order of the matrix M.

(ii) The matrix M.

b. Solve the following inequation. [3]

$$\frac{-x}{3} - 4 \leq \frac{x}{2} - \frac{7}{3} < \frac{-7}{6}, x \in R$$

Represent the solution set on a number line.

c. The first term, the last term and the common difference of an arithmetic progression are 98, 1001 and 7 respectively. Find the following for the given arithmetic progression: [4]

(i) Number of terms, 'n'

(ii) Sum of the 'n' terms.

QUESTION: (7)

a. An observer standing on the top of a lighthouse 150 m above the sea level watches a ship sailing away. As he observes, the angle of depression of the ship changes from 50° to 30°. Determine the distance travelled by the ship during the period of observation. Give your answer correct to the nearest metre. [5]

b. The daily wages of 160 workers in a building project are given: [5]

Wages (Rs)	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
No. of workers	12	20	30	38	24	16	12	8

Using graph paper, draw an ogive for the above distribution. Use your ogive to estimate:

(i) The median wage of workers

(ii) The percentage of workers who earn more than Rs 55 a day.

QUESTION: (8)

a. Mr. Sharma receives an annual income of Rs.900 in buying Rs.50 shares selling at Rs.80. If the dividend declared is 20%, find the [3]

(i) Amount invested by Mr. Sharma

(ii) Percentage return on this investment

b. A cone and a sphere having the same radius are melted and recast into a cylinder. The radius and height of the cone are 3 cm and 12 cm respectively. If the radius of the cylinder so formed is 2 cm, find the height of the cylinder. [3]

c. Use remainder theorem to factorise $6x^3 + 25x^2 + 31x + 10$ completely. [4]

QUESTION: (9)

a. Prove the following: [3]

$$\frac{cos^3A+sin^3A}{\cos A+\sin A} + \frac{cos^3A-sin^3A}{\cos A-\sin A} = 2$$

b. A box contains 7 red, 3 white and 2 black balls. When a ball is picked up from the box, find the probability that: [3]

(i) It is red

(ii) It is not white

(iii) It is black

c. A shopkeeper buys a certain number of books for Rs 360. If the cost of per book was Rs 5 less, the number of books that could be bought for Rs 360 would be one more. Taking the original cost of each book to be Rs x, write an equation in x and solve it. [4]

QUESTION: (10)

a. Construct a triangle BCP, given BC = 5.5 cm, BP = 4.5 cm and $\angle PBC = 45°$. [3]

(i) Complete the rectangle ABCD such that P is equidistant from AB and BC and P is equidistant from C and D.

(ii) Measure and record the length of AB.

b. The 1st and the 8th terms of a GP are 4 and 512 respectively. Find: [3]

(i) The common ratio

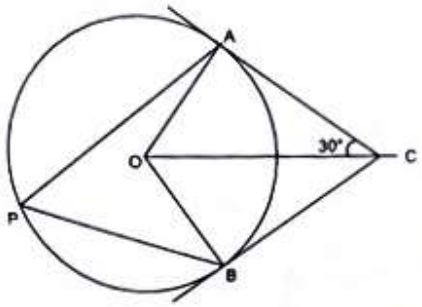
(ii) The sum of its first 5 terms.

c. In the given figure, O is the centre of the circle. Tangents at A and B meet at C. If $\angle ACO = 30°$, find: [4]

(i) $\angle BCO$

(ii) $\angle AOB$

(iii) $\angle APB$



SBDBHS

Test 2

First Preliminary Examination
Subject: Mathematics

Name of the Candidate:	
Grade: 10 ICSE	**Div: C**
	Duration: 3Hours
Maximum Marks: 80	**Roll No:**

Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during first 15 minutes.
This time is to be spent in reading the question paper.
The time given at the head of this paper is the time allowed for writing the answers.

Attempt all questions from Section A and any four questions from Section B.
All working, including rough work, must be clearly shown and must be done on the same sheet as the rest of the answer.
Omission of essential work will result in loss of marks.
The intended marks for questions or parts of the questions are given in brackets [ ]
A mathematical table is provided.
This document consists of 7 printed pages and 1 blank page.

Section A
(Attempt all questions from this section.)

Question 1:Choose the correct answers to the questions from the given options:
(Do not copy the questions; write the correct answer only) **[15]**

i) For an intrastate sale, the GST paid by a dealer to the central government is ₹ 120 if the mark price of the article is ₹ 2000, the GST is:

a. 6 %
b. 10 %
c. 12 %
d. 16.67 %

ii) It is deposited ₹ 400 per month for 15 months in Federal Bank's recurring deposit account. If the bank pays interest at the rate of 10% per annum, then the interest earned by Naresh during this period is:

a. ₹ 350
b. ₹ 250
c. ₹ 400
d. ₹ 300

iii) If $\begin{bmatrix} 2 & 0 \\ -1 & 7 \end{bmatrix}$ then A^2 is:

a. $\begin{bmatrix} 4 & 0 \\ 1 & 49 \end{bmatrix}$

b. $\begin{bmatrix} 4 & 0 \\ -9 & 49 \end{bmatrix}$

c. $\begin{bmatrix} 4 & 0 \\ 9 & 49 \end{bmatrix}$

d. $\begin{bmatrix} 4 & 9 \\ -9 & 49 \end{bmatrix}$

iv) Event A: The sun will rise from the east tomorrow.
Event B: It will rain on Monday.
Event C: February has 29 days in a leap year.
Which of the above events (s) has a probability equal to 1?

a. All events A, B and C.
b. Both events A and B.
c. Both events B and C.
d. Both events A and C.

v) The common difference of AP $\frac{1}{p}, \frac{1-p}{p}, \frac{1-2p}{p}, \ldots\ldots$ is:

a. -1
b. 1
c. $\frac{1}{p}$
d. $-\frac{1}{p}$

vi) If $(x-2)$ is a factor of $2x^3 - x^2 - px - 2$ then the value of p is:

a. 6
b. 4
c. 5
d. 8

vii) Statement 1: $Sin^2\theta + Cos^2\theta = 1$
Statement 2: $Cosec^2\theta + Cot^2\theta = 1$
Which of the following is valid:
a. Only 1
b. Only 2
c. Both 1 and 2
d. Neither 1 nor 2

viii) The roots of the quadratic equation $px^2 - qx + r = 0$ are real and equal if:
a. $p^2 = 4qr$
b. $q^2 = 4pr$
c. $-q^2 = 4pr$
d. $p^2 > 4qr$

ix) The coordinates of the vertices of ΔABC are respectively (-4,-2), (6, 2) and (4,6). The centroid G of ΔABC is :
a. (2,2)
b. (2,3)
c. (3,3)
d. (0,-1)

x) The solution set for the given inequation $-8 \leq 2x < 8$, $x \in w$ is
a. {-4,-3,-2,-1,0,1,2,3,4}
b. {-4,-3,-2,-1}
c. {0,1,2,3}
d. {-8,-7,-6,-5,-4,-3,-2,-1,0,1,2,3,4,5,6,7,8}

xi) If the first term of a GP is 10 and the common ratio is 4 then the 5th term is
a. 10240
b. 40960
c. 5120
d. 2560

xii) The product of two consecutive even whole numbers is 24, then the numbers are:
a. -8 and -3
b. 3and 8
c. -4 and -6
d. 4and 6

xiii) Ram buys 300 ten-rupee shares at 15rs and receives a dividend of 7%. The amount of investment is :

a. 3000rs
b. 3500rs
c. 4000rs
d. 4500rs

xiv) A lighthouse is 80m high. The angle of elevation of its top from a point 80m away from its foot along the same horizontal line is

a. 60
b. 45
c. 30
d. 90

xv) The roots of the quadratic equation x^2 -1 =0

a. 0,0
b. 1,1
c. -1,-1
d. +1, -1

Question 2:

i) $-11, -7, -3$ ... .49,53 are the terms of a progression. Answer the following: **[4]**

a. What is the type of progression?
b. How many terms are there in all?
c. Calculate the value of the middle-most term.

ii) Given $\begin{bmatrix} x & 3 \\ y & 3 \end{bmatrix}$ If $A^2 = 3I$. I is the identity matrix of order two. Find x and y. **[4]**

iii) Mr. Roy went to a departmental store and bought the following items from store:

Sr. No.	Items	Price	Quantity	Rate of GST
A	Walnut	750 per packet	2 Packets	5 %
B	Atta	40 per kg	20 Kg	0 %
C	Butter	200 per packet	5 Packets	12 %

[4]

Find the:

a. Total tax paid to the central government
b. Total GST paid.
c. Total bill amount including GST.

Question 3:

i) Prove that:

$$\frac{Sin\ A}{1+Cot\ A} - \frac{Cos\ A}{1+tan\ A} = Sin\ A - Cos\ A$$ [4]

ii) Using properties of proportion, find $x : y$

Given: $\frac{x^2+2x}{2x+4} = \frac{y^2+3y}{3y+9}$ [5]

iii) The difference of the squares of two natural numbers is 84. The square of the larger number is 25 times the smaller number. Find the numbers. [4]

Section B

(Attempt any four questions from this section)

Question 4:

i) Solve the quadratic equation $x^2 - 3(x+3) = 0$. Give your answer correct to two significant figures. [3]

ii) All the following inequation and represent it on a real number line:

$$-2 < \frac{1}{2} - \frac{3x+1}{4} \leq 3\frac{5}{8}; \qquad x \in R$$ [3]

iii) Prove :

$(\sin\theta + \cos\theta)(\tan\theta + \cot\theta) = \sec\theta + \text{cosec}\,\theta$ [4]

Question 5:

i) Rekha opened a recurring deposit account for 20 months. The rate of interest is 9 % per annum and Rekha receives ₹ 441 as interest at the time of maturity. Find the amount Rekha deposited each month. [3]

ii) The 1st and the 8th term of a GP are 4 and 512, respectively. Find: [3]

a. Common ratio.

b. The sum of its first 5 terms.

iii) If the line joining the points A (4, –5) and B (4, 5) is divided by the point P such that $\frac{AP}{AB} = \frac{2}{5}$ then find the coordinates of P. [4]

Question 6:

i) From the top of a tower 100m high, a man observes the angle of depression of two ships, A and B on opposite sides of the tower as $45°$ and $38°$ respectively. If the foot of the tower and the ships are in the same horizontal line, find the distance between the two ships A and B to the nearest meter. [3]

$(\tan 38° = 0.7813)$

ii) A man bought 200 shares each of face value of ₹10 per share at ₹12 per share. At the end of the year the company from which he bought the shares declares a dividend of 15%. Calculate: [4]

a. The amount of money invested by the man.
b. The amount of dividend he received.
c. The percentage return on his outlay.

iii) If $(x + 2)$ and $(x + 3)$ are factors of $x^3 + ax + b$. Find the value of a and b. [3]

Question 7:

i) An integer is chosen at random from 1 to 50. Find the probability that the number is: [5]

a. Divisible by 5
b. a perfect cube
c. a prime number
d. multiple of 15

ii) Prove that: $(1 + Cot\ A - Cosec\ A)(1 + \tan A + Sec\ A) = 2$ [5]

Question 8:

i) If x, y and z are in continued proportion, prove that: [3]

$$\frac{x}{y^2.z^2} + \frac{y}{z^2.x^2} + \frac{z}{x^2.y^2} = \frac{1}{x^3} + \frac{1}{y^3} + \frac{1}{z^3}$$

ii) Use factor theorem to factorise the following polynomial completely: [4]

$6x^3 + 17x^2 + 4x - 12$

iii) Sara opened a recurring deposit account with a nationalised bank for a period of two years. If the bank pays interest at the rate of 6% per annum and the monthly instalment is ₹ 1,000, find the: [3]

a. Interest earned in 2 years.
b. Matured value.

Question 9:

i) Marks obtained by hundred students in a mathematics test are given below: **[3]**

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
No. of Students	3	7	12	17	23	14	9	6	5	4

Draw an ogive for the given distribution on a graph sheet.

ii) If $A = \begin{bmatrix} 2 & 1 \\ 0 & -2 \end{bmatrix}, B = \begin{bmatrix} 4 & 1 \\ -3 & -2 \end{bmatrix}$ and $C = \begin{bmatrix} -3 & 2 \\ -1 & 4 \end{bmatrix}$ then find: $A^2 + AC - 5B$ **[4]**

iii) Boy is twice as old as his sister. Four years hence. The product of their ages (in years) will be 160. Find their present ages. **[3]**

Question 10:

i) Divide 216 into three parts, which are in AP, and the product of the two smaller parts is 5040. **[3]**

ii) Blank Bill shows the GST rate and the marked price of articles:

Sr. No.	Item	Marked Price	Discount	Rate of Discount
A	Dry Fruits	₹ 1,200	₹ 100	12 %
B	Flour	₹ 286	Nil	5 %
C	Bakery Products	₹ 500	10 %	12 %

[4]

Find the total amount to be paid (including GST) for the above bill.

iii) Prove the following identity: $(Sin^2\theta - 1)(tan^2\theta + 1) + 1 = 0$ **[3]**

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Test 3

FIRST PRE BOARD EXAMINATION
GRADE-10
SUBJECT:- MATHEMATICS

Student's Name:_____ Roll No: _ _
Date:- Time:- 3 hrs Maximum Marks: 80

Answers to this Paper must be written on the paper provided separately. You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the question paper.
The time given at the head of this Paper is the time allowed for writing the answers.

Attempt all questions from Section A and any four questions from Section B.
The intended marks for questions or parts of questions are given in brackets [ ].

Section A [40 marks]
(Attempt all questions from this section)

Question 1

Choose the correct answers to the following questions from the given options: [15]

i) If $x\{-3, -1, 0, 1, 3, 5\}$,then solution set of an inequation $3x - 2 \leq 8$ is

a) $\{-3, -1, 1, 3\}$ b) $\{-3, -1, 0, 1, 3\}$
c) $\{-3, -2, -1, 0, 1, 2, 3\}$ d) $\{-3. -2. -1, 0, 1, 2\}$

ii) Given that $\begin{bmatrix} 2 \\ 3 \end{bmatrix} M = \begin{bmatrix} 4 & -6 \\ 6 & -9 \end{bmatrix}$ where M is a matrix, state the order of M

a) 1x2 b) 2x1 c) 1x1 d) 2x2

iii) Anuja has a cumulative bank account and deposits ₹200 per month for a period of 2 years. If she receives ₹5300 at the time of maturity, then rate of interest per annum is

a) 5% b) 6% c) 8% d) 10%

iv) A ,B and C are three dealers in the GST chain. A buys certain goods at ₹ 2000 and sells it to B at the profit of ₹ 500 , in turn B sells the same goods to C at the profit of ₹ 1000. Taking the rate of GST at 18%,Find output tax payable by dealer B.

a) ₹ 450 b) ₹ 630 c) ₹ 90 d)₹ 180

v) The sides AB and DC of a cyclic quadrilateral ABCD are produced to meet at P; the sides AD and BC are produced to meet at Q. If $\angle ADC = 85°$ and $\angle BPC = 40°$, then $\angle CQD$ =?

a) 95° b) 55°

c) 85° d) 30°



vi) Assertion(A): Roots of the equation $x^2 + 3x + 4 = 0$ are imaginary

Reason(R) : If for the quadratic equation $ax^2 + bx + c = 0, b \neq 0, b^2 - 4ac < 0$, then its roots are imaginary.

a) Both (A) and (R) are true and(R)is the correct explanation for A

b) Both (A) and (R) are true and(R) is not the correct explanation for A

c) (A) is true but (R) is false.

d) (A) is false but (R) is true.

vii) In what ratio does the point p(-4,1)divide the line segment joining A(2,-2) and B (-14,6)?

a) 5:3 b) 3:5 c)1:3 d)3:7

viii)Locus of the centres of the circle passing through two fixed points A and B is:

a) a line parallel to line segment AB.

b) the bisector of line segment AB.

c) perpendicular to line segment AB.

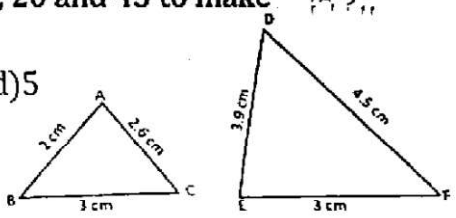
d) perpendicular bisector of line segment AB.

ix) A box contains 50 cards numbered 1 to 50. If one card is drawn from the bag at random, then probability that the number on the card is a perfect square is

a) $\frac{1}{10}$ b) $\frac{4}{25}$ c) $\frac{7}{50}$ d) $\frac{3}{25}$

x) What least number must be added to each of the numbers 6,15, 20 and 43 to make them proportional?

a) 1 b) 3 c) -3 d)5



xi) ΔABC and ΔEFD are similar by axiom

a) SAS b)AAA c) SSS d) AAS

xii) If a polynomial f(x) is divided by a linear divisor $(x - a)$,then the remainder is

a) f(x) b) f(0) c) f(a) d) f(-a)

xiii) The sum of the deviations from the _____ is always zero.

a) Mean b) Median c) Mode d) None of the above.

xiv) Which of the following points is invariant with respect to line x = - 2?

a) (2,7) b)(-2,7) c)(2,-2) d)(7,-2)

xv) G.P : $\frac{2}{9}, \frac{1}{3}, \frac{1}{2}$.....

Assertion(A) : 5th term of the given GP is $1\frac{1}{8}$.

Reason(R): If for a GP the first term is a, the common ratio is r and number of terms= n, then sum of first n terms $S_n = \frac{a(r^n-1)}{r-1}$ for all r.

a) Both (A) and (R) are true and(R)is the correct explanation for A
b) Both (A) and (R) are true and(R) is not the correct explanation for A
c) (A) is true but (R) is false.
d) (A) is false but (R) is true.

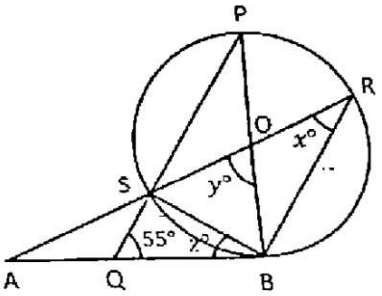
Question 2

a) Find the value of m if $(x + m)$ is a factor of $2x^2 + 2mx + 5x + 10$. For the value of m, factorise $2x^2 + 2mx + 5x + 10$ completely. [4]

b) Joseph has a recurring deposit account in a bank for 3 years at 10% p.a. simple interest. If he gets ₹16,650 as interest at the time of maturity, find his monthly deposit and the maturity value. [4]

c) In the given figure, O is the centre of the circle and AB is a tangent to the circle at B. If $\angle PQB = 55°$.

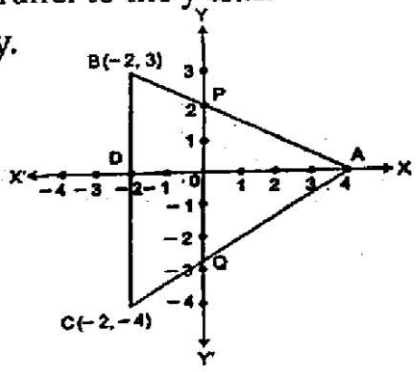
i) find the value of angles x,y and z
ii) prove that RB is parallel to PQ. [4]



Question 3

a) In the figure, given below, ABC is a triangle and BC is parallel to the y axis. AB and AC intersect the y-axis at P and Q respectively.

i) Write the coordinates of A.
ii) Find the length of AB and AC
iii) Find the ratio in which Q divides AC.
iv) Find the equation of line AC. [4]



b) Divide ₹1,870 into 3 parts in such a way that half of the first part, one third of the second part and one-sixth of the third part are all equal [4]

c) Use ruler and compass to answer this question. Construct $\angle ABC = 90$, where AB=6cm, BC= 8cm.

i) Construct the Locus of points equidistant from B and C.

ii) Construct the locus of points equidistant from A and B.

iii) Mark the point which satisfies both the conditions(i) and (ii) as O. Construct the Locus of points keeping a fixed distance OA from fixed point O.

iv) Construct the Locus of points which are equidistant from BA and BC. [5]

Section B [40 marks]
(Attempt any four out of seven questions)

Question 4

a) In a bundle of 50 shirts, 44 are good, 4 have Minor defects and 2 have major defects. What is the probability that:

i) It is acceptable to a trader who accepts only a good shirt ?

ii) It is acceptable to a trader who rejects only a shirt with major defects ? [3]

b) Find the smallest value of x for the equation $x - 3(2 + x) < 2(3x - 1)$,When:

i) $x \in W$(whole numbers)

ii) $x \in N$(natural numbers)

iii) $x \in I$(integers) [3]

c) On a map, drawn to a scale of 1:20000, a rectangular plot of land ABCD has AB=24 cm and BC=32cm. Calculate:

i) the diagonal distance of the plot in kilometre.

ii) the area of the plot in sq. km. [4]

Question 5

a) If the third and the 9th terms of an A.P. are 4 and -8 respectively, find which term is 0? [3]

b) Given $A = \begin{bmatrix} 2 & 0 \\ -1 & 7 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $A^2 = 9A + mI$. Find m. [3]

c) An aeroplane, at an altitude of 250m, observes the angles of depression of two boats on the opposite banks of a river to be 45° and 60° respectively. If the boats are on the opposite side of the aeroplane, find the width of the river. Write the answer correct to the nearest Whole number. [4]

Question 6

a) A company with 500 shares of nominal value of ₹120 declares an annual dividend of 15%. Calculate:

i) the total amount of dividend paid by the company.

ii) annual income of Mr Gulati, if he holds 200 shares.

iii) if the return percent of Mr Gulati from his shares is 10%, find the market value of each share. [3]

b) In the given figure, TP and TQ are two tangents to the circle with Centre O, touching the circle at points at A and C respectively. If $\angle BCQ = 55°$, $\angle BAP = 60°$ find:

i) $\angle OBA$

ii) $\angle OBC$

iii) $\angle AOC$

[3]

c) Two water taps can fill an empty tank in $9\frac{3}{8}$ hours. The tap of larger diameter takes 10 hours less than the smaller one to fill the empty tank separately. Find the time in which each tap can separately fill the tank. [4]

Question 7

a) Use a graph for this question. Draw an ogive for the given distribution. from the graph determine:

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
No. of students	5	10	14	21	25	34	36	27	16	12

i) the median

ii) the number of students scoring about 65 marks

iii) if 10 students qualify for merits scholarship, find the minimum marks required to qualify.

iv) the number of students who did not pass if the past percentage was 35. [5]

b) Refer to the given bill.

A customer paid ₹2000(rounded off to the nearest ₹10) to clear the bill.

Note: 5% discount is applicable on an article if 10 or more such articles are purchased.

BILL			
Article	M.P(₹)	Quantity	G.S.T
A	190	6	12%
B	50	12	18%

Check whether the total amount paid by the customer is correct or not? Justify your answer with necessary working. [5]

Question 8

a) The origin O, B (- 6, 9) and C (12, - 3) are vertices of triangle OBC. Point P divides OB in the ratio 1:2 and point Q divides OC in the ratio 1:2. Find the coordinates of point P and Q. Also show that $PQ = \frac{1}{3}BC$. [3]

b) There are three positive numbers in a geometric progression such that:
   i) their product is 3375.
   ii) the result of the product of first and second numbers added to the product of second and third number is 750. Find the numbers. [3]

c) Solve for x, using the properties of proportionality: $\frac{1+x+x^2}{1-x+x^2} = \frac{62(1+x)}{63(1-x)}$. [4]

Question 9

a) In triangle ABC, ∠ABC= 90°, side AB= 6cm, side BC=7.2cm and BD is perpendicular to side AC. Draw a circumcircle of triangle BDC and then state the length of the radius of this circumcircle drawn. [3]

b) Find the mode and modal class of the following distribution by drawing a histogram. (HINT: Mid value is given) [3]

Mid-value	28	34	40	46	52
Frequency	10	8	24	16	8

c) Use graph paper for this question. Take 2cm=1 unit along both the axes.
   i) Plot points A (2, 3), B(2, 2) and C (3, 2).
   ii) Reflect A,B and C about x-axis and name them as A' ,B' and C' respectively.
   iii) Reflect A, B, C ,A', B' and C' about y-axis and name them as A'' ,B'', C'', A''', B''' and C''' respectively.
   iv) Join A, B, C, C'', B'', A'', A''', B''', C''', C' ,B', A' and A to make a closed figure, give a geometrical name to the figure. [4]

Question 10

a) **The weights of 50 apples were recorded as given below. Calculate the mean weight to the nearest gram, by the step deviation method.** **[3]**

Weight in gram	80-85	85-90	90-95	95-100	100-105	105-110	110-115
No. of apples	5	8	10	12	8	4	3

b) Find the equation of lines which passes through the point (-2, 3) and are equally inclined to the co-ordinate axes. [3]

c) If $4\cos^2 A - 3 = 0$ and $0° \leq A \leq 90°$, then prove that:

   i) $\sin 3A = 3\sin A - 4\sin^3 A$

   ii) $\cos 3A = 4\cos^3 A - 3\cos A$ [4]

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Test 4

Preliminary Examination – 1 (2025-26)

Subject: Mathematics **Grade:10**

Time : 3 Hour **Marks : 80**

Instructions :

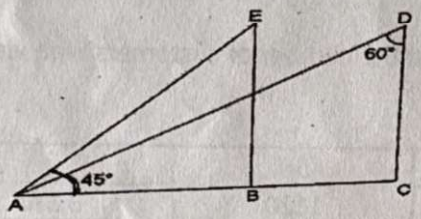
Answers to this paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the question paper.
The time given at the head of this paper is the time allowed for writing the answer.
Attempt all questions from Section A and any four from Section B.
The intended marks for questions or parts of questions are given in brackets [ ].

Section A
(Attempt all questions from this section)

Question 1 [15]

Choose the correct answers to the questions from the given options.
(Do not copy the question, write the correct answers only)

1) The angles of depression from the observing positions D and E of the object at A are
(a) 30^0 , 45^0 (b) 60^0 , 45^0 (c) 30^0 , 60^0 (d) None of these



2) The first four terms of an A.P. whose first term is -2 and the common difference is -2 are
(a) $-2, 0, 2, 4$ (b) $-2, -4, -6, -8$
(c) $-2, 4, -8, 16$ (d) $-2, -4, -8, -16$

3) When $p(x) = x^3 + ax^2 + 2x + a$ is divided by $(x + a)$, the remainder is
(a) 0 (b) $-a$ (c) a (d) 2a

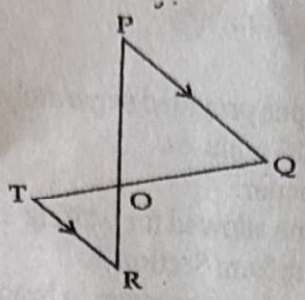
4) Assertion (A) : $\left(\frac{1+\cos A}{\sin A}\right)^2 = \frac{1+\cos A}{1-\cos A}$
Reason (R) : $sin^2A = cos^2A - 1$
(a) (A) is true and (R) is false
(b) (A) is false and (R) is true
(c) Both (A) and (R) are true and (R) is the correct explanation of (A).
(d) Both (A) and (R) are true and (R) is not the correct explanation of (A).

5) Ram opened a recurring deposit account in a bank for 2 years. He deposits Rs 1000 every month and receives Rs 25500 on maturity. The interest he earned in 2 years is
(a) Rs 13500 (b) Rs 3000 (c) Rs 24000 (d) Rs 1500

6) In the given figure PQ is parallel to TR, then by using condition of similarity

(a) $\frac{PQ}{RT} = \frac{OP}{OT} = \frac{OQ}{OR}$ (b) $\frac{PQ}{RT} = \frac{OP}{OR} = \frac{OQ}{OT}$

(c) $\frac{PQ}{RT} = \frac{OR}{OP} = \frac{OQ}{OT}$ (d) $\frac{PQ}{RT} = \frac{OP}{OR} = \frac{OT}{OQ}$



7) The reflection of the point M(1, −3) in the line y = −1 is

(a) (−3, −2) (b) (1, −4) (c) (1, 4) (d) (1, 1)

8) The CGST paid by a customer to the shopkeeper for an article which is priced at Rs 500 is Rs 15. The rate of GST charged is

(a) 1.5 % (b) 3 % (c) 5 % (d) 6 %

9) The value/s of k for which the quadratic equation $2x^2 - kx + k = 0$ has equal roots is/are :

(a) 0 only (b) 4, 0 (c) 8 only (d) 0, 8

10) The profit in rupees in a local restaurant and the number of customers who visited the restaurant are given below :

No. of customers		14	56	x	282
Profit in Rs		280	1120	3200	y

(a) x = 160 , y = 5640 (b) x = 16 , y = 546

(c) x = 16 , y = 564 (d) x = 160 , y = 5460

11) The inclination of the line $x - y\sqrt{3} + 2\sqrt{3} = 0$ is

(a) 0^0 (b) 30^0 (c) 60^0 (d) 90^0

12) If $9 \geq 14 - 5x > 3x - 5$; x ∈ W , then the solution set

(a) {0, 1, 2} (b) { 1, 2, 3 } (c) { 1, 2} (d) { 0, 1, 2, 3}

13) Assertion (A) : If the radius of a cone is halved and volume is not changed, then height remains same.

Reason (R) : If the radius of a cone is halved and volume is not changed then height must become four times of the original height.

(a) (A) is true and (R) is false

(b) (A) is false and (R) is true

(c) Both (A) and (R) are true and (R) is the correct explanation of (A).

(d) Both (A) and (R) are true and (R) is not the correct explanation of (A).

14) Rs 30 shares in a company are sold at Rs 37.50. The cost of 100 shares is

(a) Rs 3000 (b) Rs 375 (c) Rs 3750 (d) Rs 37500

15) If $A = [-1 \quad 2]$ and $B = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ then matrix BA is equal to

(a) $\begin{bmatrix} -3 & 6 \\ -4 & 8 \end{bmatrix}$ (b) $\begin{bmatrix} -3 & -4 \\ 6 & 8 \end{bmatrix}$ (c) $\begin{bmatrix} -3 \\ 8 \end{bmatrix}$ (d) $[-3 \quad 8]$

Question 2

(i) The tenth term of a G.P. is 8 times its 13th term. The first term is 3. [4]

(a) Write the expression for 10th and 13th term.

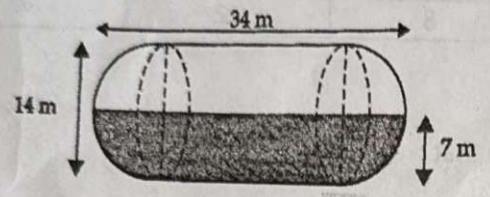
(b) Find the common ratio.

(c) Find the sum upto infinite terms.

(ii) Shown below is a horizontal water tank composed of a cylinder and two hemispheres. [4]
The tank is filled upto a height of 7 m. Find

(a) the surface area of the tank in contact with the water

(b) the quantity of water that can be stored inside the horizontal water tank.



(iii) Two people standing on the same side of a tower in a straight line with it, measures the angles of elevation of the top of the tower as 25^0 and 50^0 respectively. If the height of the tower is 70m, find the distance between the two people. [4]

Question 3

(i) If $(x-2)$ is a factor of $2x^3 - x^2 - px - 2$. [4]

(a) Find the value of p.

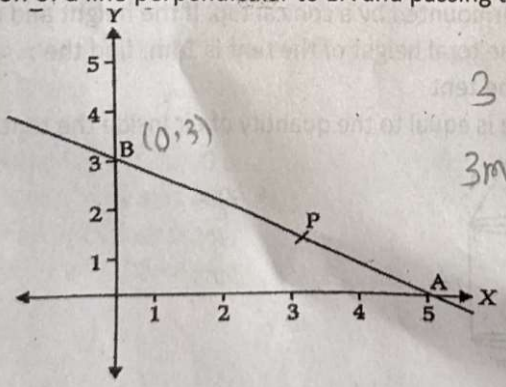
(b) with the value of p, factorise the above expression completely.

(ii) From the given figure, [4]

(a) Write down the co-ordinates of A and B. A (5,0)

(b) Find the ratio in which $P(3, \frac{6}{5})$ divides the line BA.

(c) Find the equation of a line perpendicular to BA and passing through point P.



$3 = \frac{mx_2 + nx_1}{m+n}$; my_2

$3m + 3n = \frac{0+5}{2}$

$3+3n = 5$

(iii) Using ruler and compass only construct $\angle ABC = 75^0$, AB = 6 cm and BC = 5 cm. [5]

(a) Construct the locus of all points equidistant from AB and BC.

(b) Construct the locus of all points equidistant from A and B.

(c) Mark the point which satisfies both the conditions (a) and (b) as P.
Hence, construct a circle with centre P and passing through A and B.

Section B
(Attempt any 4 questions from this section)

Question 4

(i) Mr Ram gets Rs 6455 at the end of 1 year at the rate of 14% per annum in a recurring deposit account. Find the monthly installment. [3]

(ii) Solve the following inequation and represent the solution set on the number-line. [3]

$$-3 + x \leq \frac{8x}{3} + 2 \leq \frac{14}{3} + 2x \; ; x \in I$$

(iii) The mean of the following distribution is 49.5. Find the value of 'a'. [4]

Class-interval	25 – 35	35 – 45	45 – 55	55 – 65	65 – 75
Frequency	6	a	8	12	4

Question 5

(i) If a, b and c are in continued proportion, then prove that : [3]

$$\frac{1}{a^3} + \frac{1}{b^3} + \frac{1}{c^3} = \frac{a}{b^2c^2} + \frac{b}{a^2c^2} + \frac{c}{a^2b^2}.$$

(ii) Ravi sells a certain number of Rs 20 shares paying 8 % dividend at Rs 18 and invested the proceeds in Rs 10 shares paying 12 % dividend at 50 % premium. If the change in his annual income is Rs 120, find the number of shares sold by Ravi. [3]

(iii) Use graph paper for this question. Take 2cm = 1 unit along both X and Y axes. [4]

(a) Plot the points A(2, 2) ; B(2, −2) ; C(0, −1) and D(0, 1)

(b) Reflect the points A and B on the y-axis and name it as E and F respectively and write their co-ordinates.

(c) Name two points which are invariant under the above reflection.

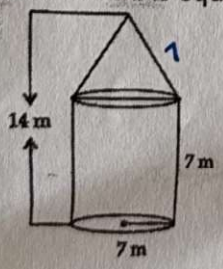
(d) Name the polygon EFCD.

Question 6

(i) A tent is in the shape of a cylinder surmounted by a conical top. If the height and radius of the cylindrical part is 7m each and the total height of the tent is 14m, find the : [5]

(a) quantity of air contained inside the tent

(b) Radius of a sphere whose volume is equal to the quantity of air inside the tent.



(ii) Use the ogive to answer the questions below : [5]

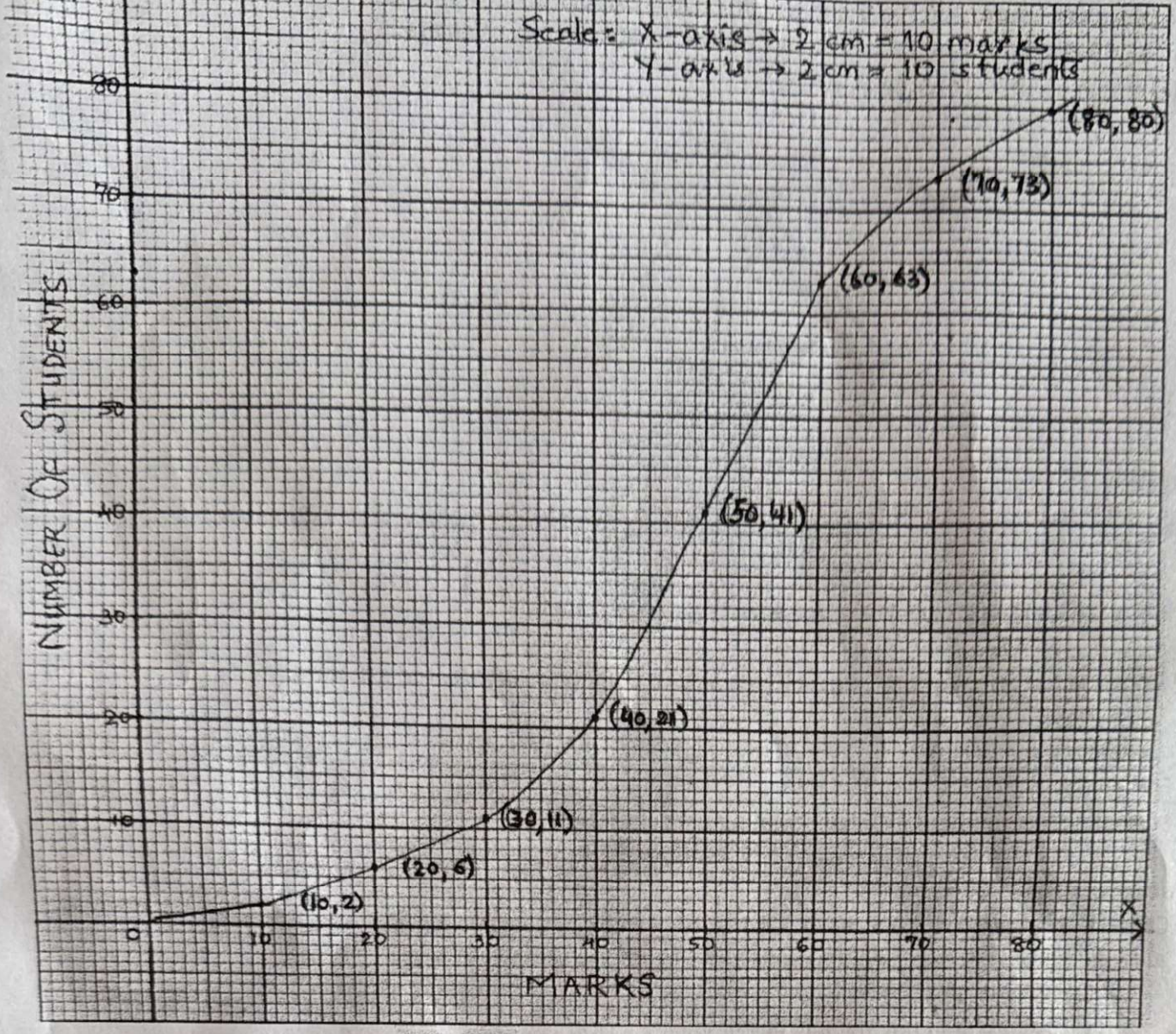
(a) the median marks

(b) Number of students scoring more than or equal to 50 but less than 70 marks.

(c) Number of students who got atleast 75 marks.

(d) The upper quartile range

(e)The total number of students.



Question 7

(i) The 2nd and 45th term of an arithmetic progression are 10 and 96 respectively. Find : [3]
(a) the common difference (b) the first term (c) the sum of the first 15 terms.

(ii) Find the equation of the line parallel to $3x - 4y + 6 = 0$ and passing through the midpoint of the join of points (2, 3) and (4, − 1). [3]

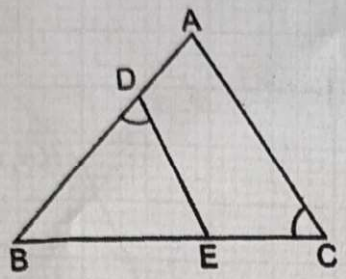
(iii) A product is sold from P to Q for Rs 8000 and then from Q to R. If the rate of tax under GST is 18 % and the profit made by Q is Rs 3000, find : [4]
(a) the GST paid by Q to State Government
(b) the GST received by Central Government
(c) the total price paid by R.

Question 8

(i) If $x = \frac{\sqrt{a+1} + \sqrt{a-1}}{\sqrt{a+1} - \sqrt{a-1}}$, then using the properties of proportion show that : [3]
$x^2 - 2ax + 1 = 0$

(ii) Solve the quadratic equation : $x^2 - 3(x + 3) = 0$. Give your answer correct to three significant figures. [4]

(iii) In the given figure, ABC is a triangle with $\angle EDB = \angle ACB$. Prove that : $\Delta ABC \sim \Delta EBD$. [3]
If BE = 6 cm , EC = 4cm, BD = 5cm and area of $\Delta EBD = 9\ cm^2$, calculate :
(a) length of AB (b) Area of ΔABC



Question 9

(i) $A = \begin{bmatrix} x & 0 \\ 1 & 2 \end{bmatrix}$; $B = \begin{bmatrix} 4 & 0 \\ y & 1 \end{bmatrix}$ Find the value of x and y, if AB = 2I, where I is a 2x2 matrix. [3]

(ii) Prove the identity : $cos^6A + sin^6A = 1 - 3cos^2A\ sin^2A$ [3]

(iii) Draw the histogram of the following frequency distribution and use it to estimate the mode. [4]

Class-interval	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
Frequency	5	15	10	5	12	8

Also, find the number of students who scored below 30.

Question 10

(i) Cards bearing number 2, 4, 6, 8, 10, 12, 14, 16, 18 and 20 are kept in a bag. A card is drawn at random from the bag. Find the probability of getting a card which is : [3]
(a) a number divisible by 4
(b) a number that is a multiple of 6 or 3
(c) an odd number.

(ii) The difference of the square of two natural numbers is 84. The square of the larger number is 25 times the smaller number. Taking x to be the smaller number, form a quadratic equation in x and solve it to find numbers. [3]

(iii) Draw the necessary diagram for this question. [4]
A pole 6m high is fixed on the top of a tower. The angle of elevation of the top of the pole observed from a point P on the ground is 60^0 and the angle of depression of the point P from the top of the tower is 45^0. Find the height of the tower and the distance of point P from the foot of the tower.

Test 5

Preliminary Examination - I

Grade: X

Time Duration: 3 Hours **Subject – Mathematics** **Maximum Marks: 80**

1. *Answers to this Paper must be written on the paper provided separately.*
2. *You will **not** be allowed to write during the first **15** minutes.*
3. *This time is to be spent in reading the question paper.*
4. ***The time given at the head of this Paper is the time allowed for writing the answers.***

5. *Attempt **all** questions from **Section A** and **any four** questions from **Section B**.*
6. ***All working, including rough work, must be clearly shown and must be done on the same sheet as the rest of the answer.***
7. ***Omission of essential working will result in loss of marks.***
8. *The intended marks for questions or parts of questions are given in brackets [ ].*
9. ***Mathematical tables and graph papers are to be provided by the school.***

SECTION A (40 Marks)

*Attempt **all** questions from this Section.*

Question 1

Choose the correct answers to the questions from the given options. [15]

(Do not copy the question, write the correct answers only)

(i) Mr. Das invests in ₹ 100, 12% shares of Company 'A' available at ₹ 60 each.
Mr. Singh invests in ₹ 50, 16% shares of Company 'B' available at ₹ 40 each.
Which of the following statements is correct?

(a) The rate of return for Mr. Das is 12%

(b) The rate of return for Mr. Singh is 10%

(c) Both Mr. Das and Mr. Singh have the same rate of return of 10%

(d) Both Mr. Das and Mr. Singh have the same rate of return of 20%

(ii) Which of the following is not a geometric progression?

(a) $\frac{1}{3}, 1, 3, 9$

(b) $\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}$

(c) $-2, 4, -8, 16$

(d) $2, 0, 4, 0, 8, 0$

ORION

(iii) A cylindrical metallic wire is stretched to double its length.

Which of the following remains constant after stretching?

(a) Curved surface area

(b) Total surface area

(c) Volume

(d) radius

(iv) In the given figure, line l_1 is a parallel to line l_2.

If line l_3 is perpendicular to line l_1,

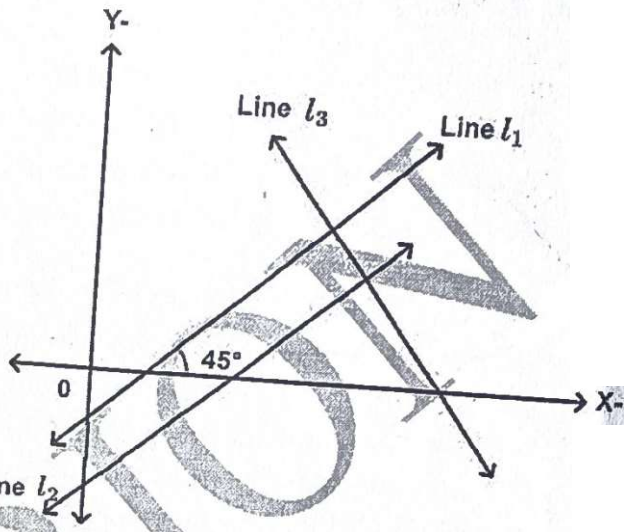
then the slopes of lines l_2 and l_3 respectively are:

(a) 1, 1

(b) -1, -1

(c) 1, -1

(d) -1, 1



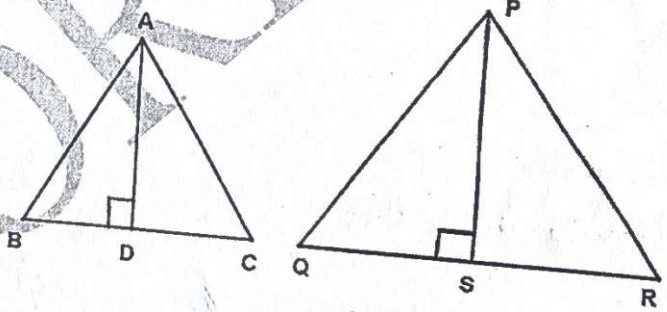
(v) In the given figure, If $\Delta ABC \sim \Delta PQR$ and $\frac{AD}{PS} = \frac{5}{11}$, find $\frac{\text{Perimeter}(\Delta ABC)}{\text{Perimeter}(\Delta PQR)}$

(a) 11 : 5

(b) 5 : 6

(c) 5 : 11

(d) 6 : 11



(vi) Locus of a moving point is _____ if it moves such that it keeps a fixed distance from a fixed point.

(a) circle

(b) line

(c) angle

(d) line segment

(vii) In the adjoining figure, O is the center of the circle, and a semicircle is drawn on OA as diameter.

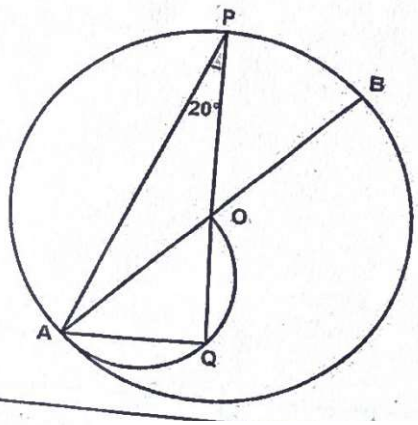
$\angle APQ = 20°$. The degree measure of $\angle OAQ$ is:

(a) 25°

(b) 40°

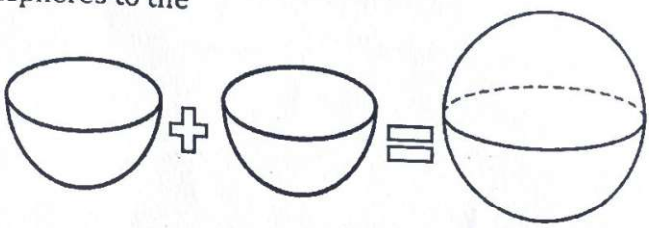
(c) 50°

(d) 65°



PMS ORION

(viii) Two identical solid hemispheres of radius 'r' are kept in contact to form a sphere. The ratio of the total surface areas of two hemispheres to the surface area of the sphere formed is:



(a) 1 : 1

(b) 3 : 2

(c) 2 : 3

(d) 2 : 1

(ix) **Statement 1:** The median of the data 5, 9, 7, 13, 10, 11, 10 is 13.

Statement 2: For a set of data with an odd number of observations, the median is the middle number.

(a) Both the statements are true.

(b) Both the statements are false.

(c) Statement 1 is true and Statement 2 is false.

(d) Statement 1 is false and Statement 2 is true.

(x) **Assertion (A):** If x is proportional to y and $x = 12$ when $y = 18$, then when $y = 9$, the value of x is 6.

Reason (R): When two quantities are directly proportional, the ratio $\frac{x}{y}$ remains constant.

(a) A is correct but R is incorrect.

(b) A is incorrect but R is correct.

(c) Both A and R are correct and R is the correct explanation of A.

(d) Both A and R are correct but R is not the correct explanation of A.

(xi) Evaluate $(\sec\theta - \tan\theta)(\sec\theta + \tan\theta)$.

(a) 0

(b) 1

(c) 2

(d) 3

(xii) If $A = [5 \quad -2]$ and $B = \begin{bmatrix} 7 \\ 11 \end{bmatrix}$, which of the following operation is/are possible?

1. A – B 2. A + B 3. AB 4. BA

(a) Only 3

(b) Both 1 and 2

(c) Only 4

(d) Both 3 and 4

PRO ORION

(xiii) $A(-3,4), B(-3,-2)$ and $C(1,-2)$ are the vertices of a right-angled triangle, right angled at B. Find the coordinates of its orthocenter:

(a) $(-3,4)$

(b) $(-3,-2)$

(c) $(1,-2)$

(d) $(-1,1)$

(xiv) If $x^2 + kx - 28 = (x - 7)(x + 4)$ for all values of x, then the value of k is

(a) -11

(b) -3

(c) 3

(d) 11

(xv) **Assertion (A):** The roots of the quadratic equation $x^2 + 12x - k = 0$ are real and different if $k > -36$.

Reason (R): For the roots of a quadratic equation $ax^2 + bx + c = 0$ to be real and distinct, the discriminant $b^2 - 4ac > 0$

(a) A is correct but R is incorrect.

(b) A is incorrect but R is correct.

(c) Both A and R are correct and R is the correct explanation of A.

(d) Both A and R are correct but R is not the correct explanation of A.

Question 2

(i) What should be added to the polynomial $x^3 + 13x^2 + 32x$ so that $x + 2$ is a factor of the resulting polynomial? Hence, find all the factors of the resultant polynomial. [4]

(ii) Calculate the ratio in which the line joining $A(-4, 2)$ and $B(3, 6)$ is divided by $P(x, 3)$. Also, find: a) x b) length of AP. [4]

(iii) Use graph paper for this question. [4]

Plot points $A(0, 3), B(4, 0), C(6, 2)$ and $D(5, 0)$.

Reflect the points as given below and write their coordinates:

(a) Reflect A in the $x - axis$ to get A'.

(b) Reflect B in the $y - axis$ to get B'.

(c) Reflect C in the $x - axis$ to get C'.

(d) Join the points A, B, A', B', A in order to form a closed figure.

Name the closed figure ABA'B'.

ORION

Question 3

(i) The sequence 2, 9, 16, ... is given. [4]

a) Determine whether the sequence is an AP or a GP. Give a reason.

b) Find the 20th term of the sequence.

c) Find the difference between the sum of its first 25 terms and the sum of its first 22 terms.

(ii) A mathematics teacher uses certain amount of terracotta clay to form different shaped solids. First, she turned it into a sphere of radius 7 cm and then she made a right circular cone with base radius 14 cm. Find the height of the cone so formed. [4]

Also, compare the total surface areas of sphere and cone so formed.

(iii) In the given figure, [5]

ABCD is cyclic quadrilateral, $\angle ADC = 135°$.

Sides BA and CD are produced to meet at P,

sides AD and BC are produced to meet at Q,

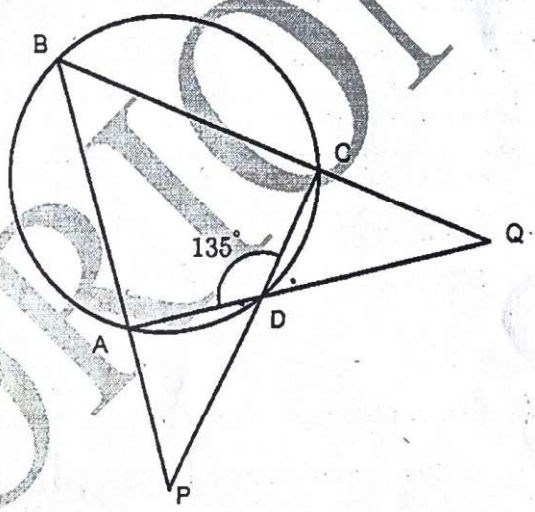
if $\angle P : \angle Q = 2 : 1$,

find:

a) $\angle P$

b) $\angle Q$

c) $\angle DCB$



SPECIMEN QUESTION

SECTION B (40 Marks)

*Attempt **any four** questions from this Section.*

Question 4

(i) Solve the following inequation and represent the solution set on a number line: [3]

$4x - 19 < \frac{3x}{5} - 2 \leq -\frac{2}{5} + x, x \in R$

(ii) A man invests a sum of money in ₹100 shares, paying 15% dividend, quoted at 20% premium. His annual dividend is ₹540. [3]

Calculate:

a) His total investment.

b) The rate of return on his investment

(iii) Prove the following trigonometric identity: [4]

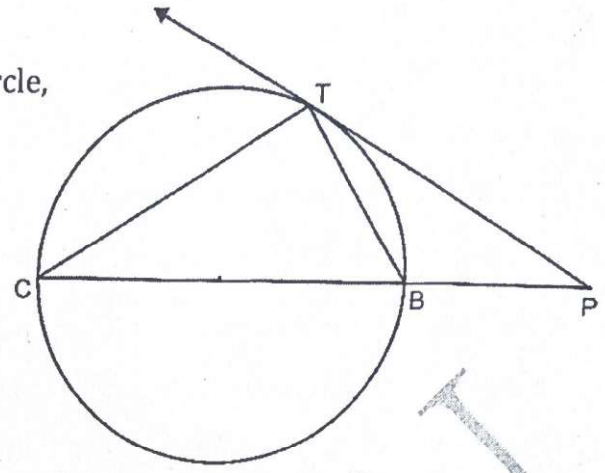
(cosec A – sin A) (sec A – cos A) (tan A + cot A) = 1.

Question 5

(i) In the figure, PB = BT and PT is tangent to the circle, then prove that:

a) △PTC is an isosceles triangle

b) PB x PC = TC^2



[3

(ii) DTDC has three types of courier service for its premium clients. An industrial house has given following orders. [3

Types of Services	Gold	Platinum
Number of Services	18	16
Cost of each service (in ₹)	200	300
Discount (%)	10%	20%
Rate of GST	18%	28%

Find:

a) Discounted cost of all services b) Total GST c) Total amount of the bill.

(iii) The following table shows the expenditure of 60 boys on books. [4]

Expenditure (in ₹)	20-25	25-30	30-35	35-40	40-45	45-50
No. of Students	4	7	23	18	6	2

Use graph sheet for this question.

a) Draw a histogram for the data using a scale of 2 cm = ₹5 on the x-axis and 2 cm = 5 students on the y-axis.

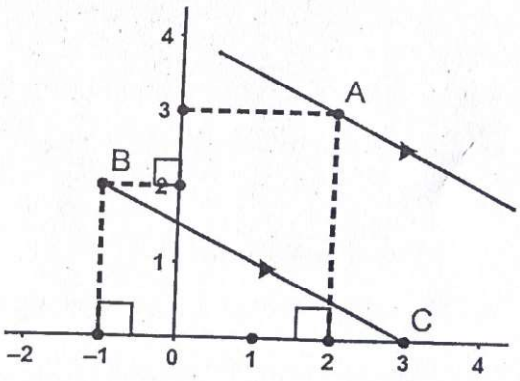
b) Find the mode of their expenditure.

Question 6

(i) In the adjoining figure, write down [3]

a) the coordinates of the points A, B and C

b) the equation of the line through A, parallel to BC.



(ii) Solve the given equation $2x^2 - 9x - 6 = 0$ and express your answer correct to one place of decimal. [3]

PREVIOUS

(iii) In a recurring deposit account for 2 years, the total amount deposited by Arsh is ₹14,400. [4]
If the interest earned is three-eighth of his total deposit. Find:

a) The interest he earns b) His monthly deposit c) The rate of interest.

Question 7

(i) The angle of depression of two cars A and B on the opposite side of a skyscraper of height 100 m are 42° and 54° respectively. The line joining the two cars passes through the floor of the skyscraper. Find the distance between the two cars, correct to the nearest metre. [4]

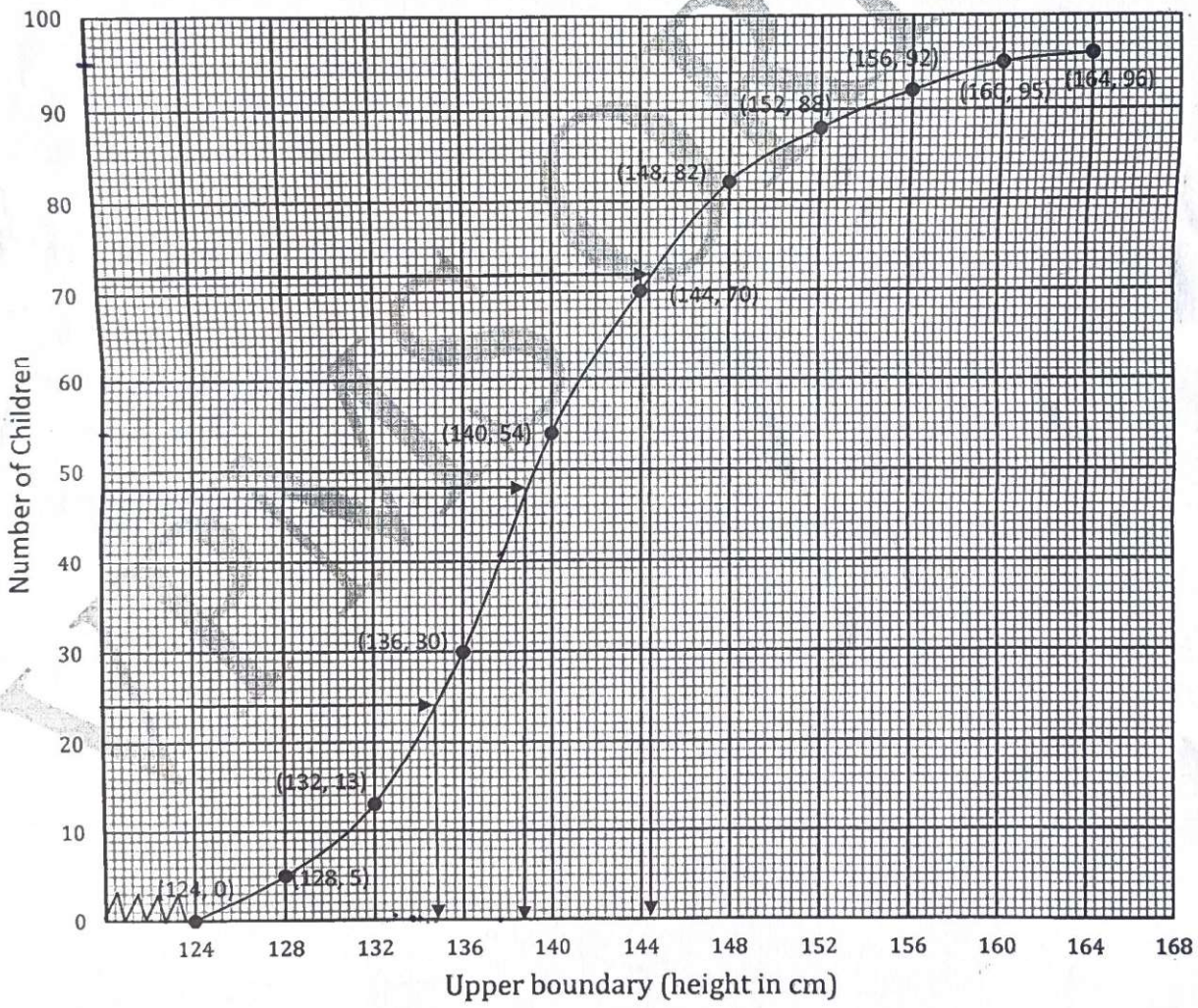
(ii) Study the graph and answer each of the following: [6]

a) Name the curve plotted b) Total number of children

c) The median height d) The lower quartile

e) The upper quartile

f) Number of children having height between 140 and 160 cm.



Question 8

(i) How many terms of the G.P. $3, \frac{3}{2}, \frac{3}{4}, \ldots$ are needed so that the sum is $\frac{3069}{512}$?

(ii) Using Componendo and Dividendo property, find a : b.

$$\frac{a^3 + 3ab^2}{3a^2b + b^3} = \frac{63}{62}$$

(iii) A toy is in the shape of a right circular cylinder with a hemisphere on one end and a cone on the other. The height and radius of the cylindrical part are 13 cm and 5 cm respectively. The radii of the hemispherical and conical parts are the same as that of the cylinder. If the height and slant height of the cone are 12 cm and 13 cm respectively, calculate the total surface area of the toy.

Question 9

(i) Use the step-deviation method to find the mean of the following distribution.

Monthly Wages (in ₹1000)	90-110	110-130	130-150	150-170	170-190
Number of Men	4	6	4	8	18

(ii) Each of the letter of the word 'HOUSEWARMING' is written on separate cards and placed in a bag.

If a card is drawn at random after shuffling,

find the probability that the letter on the card is:

a) A vowel

b) One of the letters of the word 'SEWING'

c) Not a letter from the word 'WEAR'

(iii) In the figure given below (not drawn to the scale),

AD ∥ GE ∥ BC,

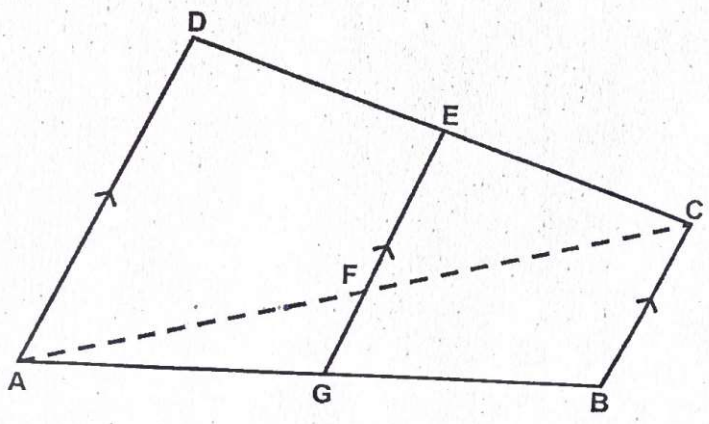
DE = 18 cm, EC = 3 cm, AD = 35 cm.

Find:

a) AF : FC

b) The length of EF

c) A(△ACD) : A(ΔFCE)



Question 10

(i) The length of a rectangular garden is 12m more than its breadth. The numerical value of its area equals four times the numerical value of its perimeter. Assuming the breadth of the garden is x metres, find its dimensions. [3]

(ii) If $A = \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix}$ and I is the 2×2 identity matrix, then find the value of 'k' if $A^2 = 8A + kI$ [3]

(iii) Using a ruler and compass only, construct a regular hexagon ABCDEF with each side equal to 4 cm. [4]

Ensure that all construction arcs and lines are clearly shown.

Then, using the same figure:

a) Draw the locus of points equidistant from any two adjacent vertices of the hexagon.

b) Draw the locus of points equidistant from any two adjacent sides of the hexagon.

c) Calculate the radius of the incircle of regular hexagon ABCDEF from above construction.

∞∞∞

VPMS ORION

Test 6

FIRST PRELIMINARY EXAMINATION
SUBJECT: MATHEMATICS

GRADE: 10

MARKS: 80
TIME: 3 HOURS

Answers to this paper must be written on the answer script provided by the school.
*You will **not** be allowed to write during the first 15 minutes.*
This time is to be spent in reading the question paper.
The time given at the head of the paper is the time allowed for writing the answers.

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*Attempt all questions from **Section A** and any four questions from **Section B**.*
All working, including rough work, must be clearly shown, and must be done on the same sheet as the rest of the answer.
Omission of essential working will result in loss of marks.
The intended marks for questions or parts of questions are given in brackets [ ].
This paper consists of 8 printed pages.

SECTION A (40 Marks)
(Attempt all questions from this section.)

Question 1

Choose the correct answers to the questions from the given options: [15]
(Do not copy the question, write the correct answers only.)

i) If $2x + 3y : 3x + 5y = 18 : 29$, then the ratio $x : y$ is:

(a) 2 : 3
(b) 3 : 5
c) 3 : 4
(d) 5 : 29

ii) For an intrastate sale the CGST paid by a dealer is ₹120. If the marked price of the article is ₹2000, the rate of GST is:

a) 6%
b) 12%
c) 10%
d) 16.67%

iii) Assertion (A) : $sec^2\theta = 1 - tan^2\theta$ is not a trigonometrical identity.

Reason (R): For an acute angle θ, the trigonometrical equation is an identity, if it is satisfied for every value of angle θ.

a) A is true, R is false.
b) A is false, R is true.
c) Both A and R are true and R is the correct explanation of A.
d) Both A and R are true and R is not the correct explanation of A.

iv) A metallic cone of height 56 cm and base radius 7 cm is melted to make small spherical balls of radius 1.4 cm. The number of balls formed is:

a) 250
b) 300
c) 350
d) 400

v) If $A = \begin{bmatrix} 5 & 7 \\ 2 & 1 \end{bmatrix}, I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $I \times A$ *is*:

a) $\begin{bmatrix} 5 & 7 \\ 2 & 1 \end{bmatrix}$
b) $\begin{bmatrix} 7 & 1 \\ 5 & 2 \end{bmatrix}$
c) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
d) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

vi) 20 tickets on which number 1 to 20 are written are mixed thoroughly and then a ticket is drawn at random out of them. The probability that the number on the drawn ticket is a multiple of 3 or 7 is:

a) $\frac{1}{5}$
b) $\frac{3}{4}$
c) $\frac{1}{4}$
d) $\frac{2}{5}$

vii) **Statement (I):** The point equidistant from all the three vertices of a triangle is the circumcentre.

Statement (II): The circumcentre is obtained by the intersection of the medians of a triangle.

a) Statement I is false but Statement II is true.
b) Statement I is true but Statement II is false.
c) Both Statement I and Statement II are true.
d) Both Statement I and Statement II are not true.

viii) Amit invested a certain sum of money in ₹100 shares paying a 7.5% dividend. The rate of return on his investment is 10%. The money invested by Amit to purchase 10 shares is:

a) ₹75
b) ₹900
c) ₹750
d) ₹1100

ix) Which term of an AP 21,42,63,84, ... *is* 252 is:

a) 12th
b) 11th
c) 13th
d) 10th

x) The vertices of a parallelogram in order are $A(1,2), B(4,y), C(x,6)$ and , $D(3,5)$. Then values of x and y are:

a) $x = 6, y = 3$

b) $x = 3, y = 6$

c) $x = 5, y = 6$

d) $x = 1, y = 4$

xi) An airplane is flying at an altitude of 1km. From a point on the ground the angle of elevation of the airplane is 60°. The following can be found using this information :

a) Height of the airplane.

b) Horizontal distance of the airplane from the observer.

c) Speed of the airplane.

d) Length of runway required for landing.

xii) The largest value of x for which, $3(x-2) \leq 6 - x$, where $x \in W$, is:

a) 3

b) 4

c) 6

d) 8

xiii) The value of $'k'$ for which the quadratic equation $2x^2 - kx + k = 0$ has equal roots is(are):

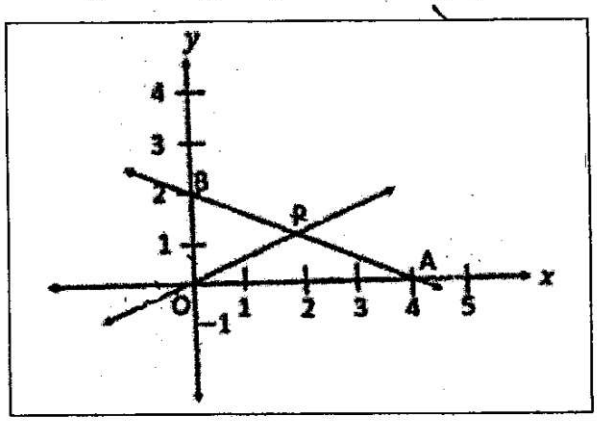
a) 0 *only*

b) 4,0

c) 8 only

d) 0,8

xiv) In the given diagram, O is the origin, P is the midpoint of AB. The equation of OP is:



a) $y = x$

b) $2y = x$

c) $y = 2x$

d) $y = -x$

xv) The students are asked to compute the mean marks for the distribution. One student divided $\sum fx$ by $\sum (fx)^2$. The type of error committed by the student is:

a) Copied wrong value of x.

b) Interchanged f and x values .

c) Omitted one frequency.

d) Chose wrong formula.

Question 2

i) Prove that: $\frac{cos^2\theta}{1-tan\theta} + \frac{sin^3\theta}{sin\theta-cos\theta} = 1 + sin\theta cos\theta$ [4]

ii) If 10,15,20,25.....are in AP. Find [4]

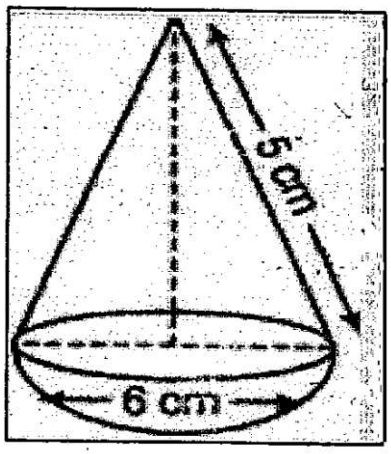
a) the $n^{th}term$ of this AP in terms of n.

b) How many terms of above AP will give the sum 3895?

iii) The given figure represents a hemisphere surmounted by a conical block of wood. The diameter of their bases is 6cm each and the slant height of the cone is 5cm.Calculate: [4]

a) The volume of the solid.

b) The total surface area of the structure. **(Take $\pi = 3.14$)**



Question 3

i) The marks obtained by 60 students in a test are recorded as follows: [4]

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70
Number of students	5	9	12	10	15	6	3

Calculate the **mean marks** of the students using **step –deviation** method.

ii) Use graph paper for this question. [4]

a) Plot $A(5,0), B\ (2,4), C(2,2), D(-4,2)$

b) Reflect B, C, D on $X - axis$ and write down their coordinates name them $B', C'\ and\ D'$ respectively.

c) $A, B, C, D, D', C', B', A'$ in order and write the name of the figure.

d) Write the coordinates of the point invariant under reflection of $Y - axis$.

[Scale: On $x\ and\ y - axis, 2\ cm\ =\ 1\ unit$]

iii) Calculate the ratio in which the line joining the points $A(-3,-1)$and $B(5,7)$ is divided by the line $x = 2$ at point P. Find the co-ordinates of P and equation of the line passing through it and perpendicular to AB. [5]

SECTION B (40 Marks)

(Attempt any 4 questions from this section.)

Question 4

i) Mr. Parekh invested ₹42,000 on ₹100 shares at a discount of ₹30 paying 8% dividend. At the end of one year he sells the shares at a premium of ₹20. Find: [3]

a) the annual dividend.

b) the profit earned including his dividend.

ii) Solve the following in-equation, write the solution set and represent it on the real number line. [3]

$$2x - 3 \leq \frac{2x+1}{3} > \frac{2x}{5}; x \in R$$

iii) Use ruler and compass [4]

Construct a ΔABC, where $AB = 6cm$ and $BC = 7.5cm$ and $\angle BAC = 75°$ hence:

a) Construct the locus of points equidistant from BA and BC.

b) Construct the locus of points equidistant from BA and AC.

c) Mark the point which satisfies the above two loci as P. Construct incircle and record the length of the radius.

Question 5

i) Ms. Reena, has a recurring deposit account for 2 years at 8% simple interest p.a. She receives ₹1520 as interest on maturity. Find: [3]

a) The monthly installment.

b) The total deposit.

c) The maturity value.

ii) The table given below shows the age of members of a society: [3]

Age (years)	25-35	35-45	45-55	55-65	65-75	75-85
No. of members	5	32	69	80	61	13

Use graph sheet for the question.

[Take $2cm = 10\ years$ along one axis and $2cm = 10\ members$ along another axis.]

a) Draw a histogram representing the above distribution

b) Estimate the modal age of the members.

iii) In the Given figure, A(0,4) and P(0,10). [4]

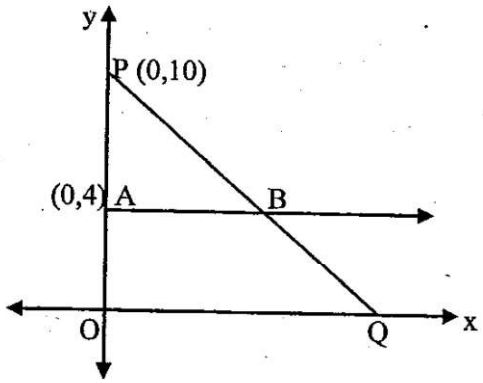
AB ∥ x-axis. PB extended meets x-axis at Q.

a) Write the equation of AB.

b) If slope of PQ= -2, find the equation of PQ.

c) Find the co-ordinates of B and Q.

d) Find the co-ordinates of the centroid of ΔPAB.



Question 6

i) If $A = \begin{bmatrix} 4 & 5 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$ and $AX = B$, Find: [3]

a) The order of matrix X.

b) Matrix X.

ii) The fifth term of a GP is 81 and its second term is 24. Find the G.P. [3]

iii) Samar went to a mobile selling shop. He bought the following items given below: [4]

Sr.no	Items	Marked price	Rate of Discount	GST%
1	Mobile	₹24000	10%	12%
2	Memory Card	₹1000	5%	18%
3	Charger	₹1200	-	18%
4	Ear Phones	₹500	-	10%

Find the: a) Total GST paid.

b) Total bill amount including GST.

Question 7

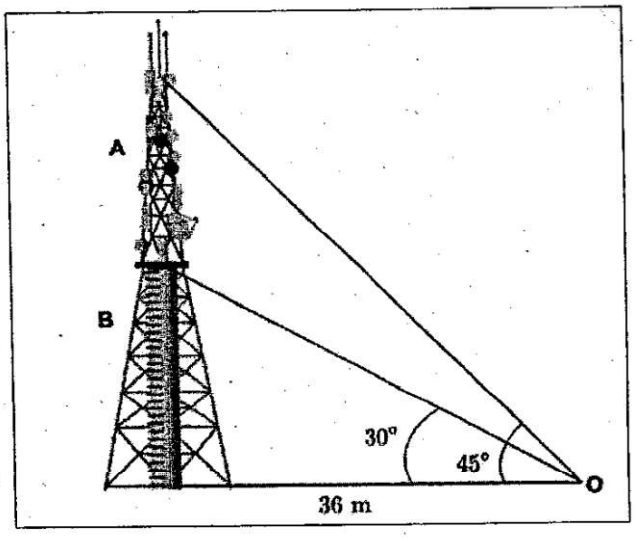
i) A radio station tower was built in two sections A and B. Tower is supported by wires from a point O. Distance between the base of the tower and point O is 36 m. From point O, the angle of elevation of the top of section B is 30° and the angle of elevation of the top of section A is 45°. [5]

a) What is the height of the section B?

b) What is the height of the section A?

c) Calculate the height of the tower section AB to the nearest meter.

d) What is the angle of depression from top of tower to point O?



ii) During a medical checkup of 60 students in a school, weights were recorded as: [5]

Weight in kg	28-30	30-32	32-34	34-36	36-38	38-40	40-42	42-44
No of students	2	4	10	13	15	9	5	2

Taking 2 $cm = 2\ kg$ along one axis and $2cm = 10\ students$ along the other axis Draw an ogive. Use graph to estimate and find the:

a) Median

b) Upper Quartile

c) Number of students whose weight is above 37 kg.

Question 8

i) Each of the letters of the word 'AUTHORIZES' is written on identical circular disc and put in a bag. They are well shuffled. If a disc is drawn at random from the bag, what is the probability that the letter is: [3]

a) a vowel?

b) one of the first 9 letters of the English alphabet which appears in the given word?

c) one of the last 9 letters of the English alphabet which appears in the given word?

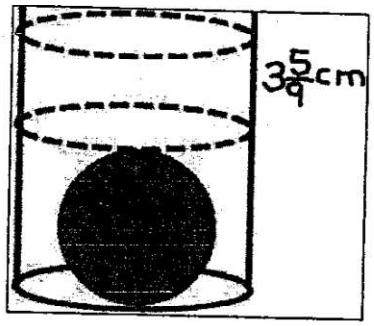
ii) Solve for x using the quadratic formula, give your answer correct to three significant figures.

$$4x^2 - 5x - 3 = 0$$ [3]

iii) A sphere of diameter 12cm is dropped in a right circular cylindrical vessel partly filled with water. If the sphere is completely submerged in water, the water level in the cylindrical vessel rises by $3\frac{5}{9}cm$. Find: [4]

a) the diameter of cylindrical vessel.

b) The capacity of the cylinder in cubic cm.
Height of the cylinder is 1.5 times the radius of the sphere.
(up to 2 places of decimal)



Question 9

i) Using factor theorem, factorize the following polynomial: [3]

$$x^3 + 10x^2 - 37x + 26$$

ii) Prove that: $\sqrt{sec^2\theta + cosec^2\theta} = tan\theta + cot\theta$ [3]

iii) The sum of four consecutive numbers in an AP is 32 and the ratio of the product of the first and the last term to the product of two middle terms is 7:15. Find the numbers. [4]

Question 10

i) The mean proportion between two numbers is 6 and their third proportion is 48. Find the two numbers. [3]

ii) $A = \begin{bmatrix} 2 & 0 \\ -1 & 7 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, find $A^2 = 9A + mI$. Find m. [3]

iii) The time taken by a person to cover 150 km was $2\frac{1}{2}$ hours more than the time taken in the return journey. If he returned at a speed of 10 km/hour more than the speed while going, find the speed per hour in each direction. [4]

Test 7

First Preliminary Examination

MATHEMATICS

Grade: 10

Max. Marks: 80

Time Allowed: 3 hours

INSTRUCTIONS: -

A. Answers to this paper must be written on the paper separately.

B. You will not be allowed to write during the first 15 minutes.

C. This time is to be spent in reading the Question Paper

D. The time given at the head of this paper is the time allowed for writing the answers.

E. The intended marks for the questions are given in brackets [ ].

F. Section I is compulsory. Attempt any four questions from Section II.

Section I (40 Marks)

(Attempt all questions from this section)

Question 1 **[15]**

Choose the correct option

) If sum of n terms of an arithmetic progression is $S_n = n^2 - n$, then the third term of the series is

a) 2
b) 4
c) 6
d) 9

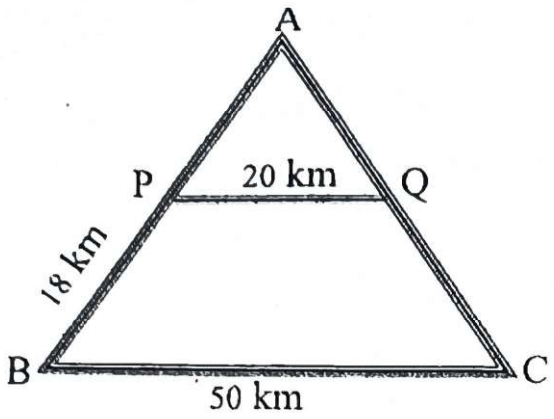
ii) The ratio in which the X-axis divides the line joining the points (-3,6) and (2,-8) is

a) 2 : 1
b) 3 : 2
c) 2 : 3
d) 3 : 4

iii) a) The maximum frequent value in a distribution is called

b) mean
c) median
d) mode
e) quartile

iv)



In the given diagram (not drawn to scale) railway stations A,B,C,P and Q are connected by straight tracks .Track PQ is parallel to BC .The time taken by a train travelling at 90 km/h to reach B from A by the shortest route is

a) 8 min
b) 12 min
c) 16.8 min
d) 20 min

v) Assertion (A) If $\sin^4 A + \sin^2 A = 1$ then $\cos^4 A + \cos^2 A = 1$
Reason (R) $1 - \sin^2 A = \cos^2 A$

a) (A) is true, (R) is false
b) (A) is false (R) is true
c) Both (A) and (R) are true and (R) is the correct reason for (A)
d) Both (A) and (R) are true and (R) is the incorrect reason for (A)

vi) If matrix $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$ and $A^2 = \begin{bmatrix} 4 & x \\ 0 & 4 \end{bmatrix}$ then the value of x is

a) 2
b) 4
c) 8
d) 10

vii) Ankit had the option of investing in company A, where 7 %, ₹100 are available at ₹120 or in a company B where 8 %, ₹1000 shares are available at ₹1620.

Assertion (A) Investing of company A is better than company B

Reason (R) The rate of income in Company A is better in company B

a) Both A and R are true, and R is the correct explanation of A

b) Both A and R are true, but R is not the correct explanation of A

c) A is false but R is true

d) Both A and R are false

viii) The sum and product of the roots of a quadratic equation are 3 and -4, the equation is

a) $x^2 + 3x - 4$
b) $x^2 - 3x + 4$
c) $x^2 + 3x + 4$
d) $x^2 - 3x - 4$

ix) If $\frac{x}{2} - 5 \leq \frac{x}{3} - 4$ and x is an odd natural number , then the solution set is

a) { -3, -1 ,1,3,5,7}

b) { -3 ,-1,1,3,5}

c) { 1,3,5,7}

d) {1,3 ,5 }

x) The equation $\sqrt{15 - 2x} = x$

Assertion (A) x = 3

Reason (R) $\sqrt{15 - 2x} = x$ implies $15 - 2x = x^2$

$x^2 + 2x - 15 = 0$

x = -5 or x= 3

a) (A) is true, (R) is false

b) (A) is false (R) is true

c) Both (A) and (R) are true and (R) is the correct reason for (A)

d) Both (A) and (R) are true and (R) is the incorrect reason for (A)

xi) 'Ms Real Paint' sold 2 tins of lustre paint and taxable value of each tin of ₹ 2800. If the rate of GST is 28%, then the amount of GST and SGST charges in the tax invoice is

a) ₹ 750

b) ₹1568

c) ₹784

d) ₹1500

xii) Assertion (A) The point (-2,8) is invariant under the reflection in the line x = -2

Reason (R) If the point has its x- coordinate O (origin), it is invariant under reflection of both the axes.

a) Both (A) and (R) are true and (R) is the correct reason for (A)

b) Both (A) and (R) are true and (R) is the incorrect reason for (A)

c) A) is true, (R) is false

d) Both (A) and (R) are true

xiii) The point of concurrence of the angular bisectors of a triangle is called the _____of the triangle

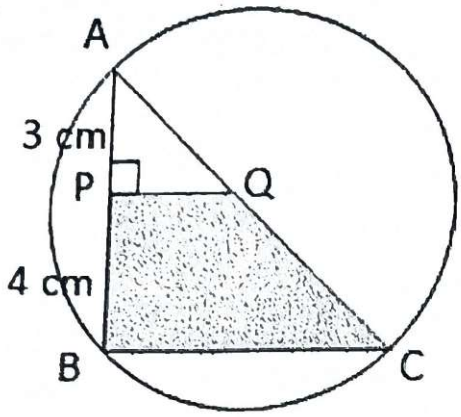
a) centroid

b) incentre

c) circumcentre

d) orthocentre

xiv)



In the figure given below, AC is a diameter of the circle. AP = 3 cm and PB = 4 cm and QP ⊥ AB. If the area of ΔAPQ is18 cm2 , then the area of shaded portion QPBC is

a) 32 cm^2

b) 49 cm^2

c) 80 cm^2

d) 98 cm^2

xv) Which of the following equations represent a line equally inclined to the axis ?

a)2x – 3y+7 = 0

b) x – y = 7

c)x = 7

d) y – 4=0

Question 2

i) Prove the following trigonometrical identities **[4]**

a)$(\sin\theta + \cos\theta)(\text{cosec}\theta - \sec\theta) = \text{cosec}\theta\sec\theta - 2\tan\theta$

b)$\frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta} = \tan\theta$

ii) Cards marked with numbers 0,1,2,3,4,.....20 are shuffled and a card is drawn at random. What is the probability that the number of the card is **[4]**

i) a prime number

ii) divisible by 3

ii) a perfect square

iii) Use a graph sheet for this question .Draw a histogram for daily earnings of 54 medical stores in the following table and hence estimate the mode for the following distribution .Take 2cm = ₹ 500 units alongside the X –axis and 2cm = 5 stores along the y- axis **[4]**

DailyEarning(₹)	4500-5000	5000-5500	5500-6000	6000-6500	6500-7000
No. of medical stores	20	14	12	5	3

Question 3

i)a) Given $A = \begin{bmatrix} 2 & -6 \\ 2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} -3 & 2 \\ 4 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$ Find the matrix X such that [4]

$$A + 2X = 2B + C$$

b) Solve the following inequality and graph the solution on the number line

$$\frac{3x+15}{5} \leq 2x + 3 \leq \frac{3x+15}{2} \text{ , } x \in R$$

ii) The horizontal distance between two towers is 120m.The angle of elevation of the top and the angle of depression of the bottom of the first tower from the top of the second tower is 30^0 and 24^0 resp. Find the height of the two towers. Give your answer correct to 3 significant figures. [4]

iii) Use graph paper for this question. [5]

A survey regarding height (in cm) of 60 boys belonging to class 10 of a school was conducted. The following data was recorded:

Height (cm)	135-140	140-145	145-150	150-155	155-160	160-165	165-170
No of boys	4	8	20	14	7	6	1

Taking 2cm = height of 10 cm along one axes and 2cm =10 boys along the other axis draw an ogive of the above distribution. Use the graph to estimate the following

a) the median

b) Lower Quartile

c) if above 158cm is considered as tall boys of the class, find the number of boys in the class who are tall.

Section II (40 Marks)

*(Attempt **any four** questions from this section)*

Question 4

i) Using componendo and dividendo ,find the value of x ,when [3]

$$\frac{x^3 + 3x}{3x^2 + 1} = \frac{14}{13}$$

ii) Prove that [3]

$$\frac{\cos^2 A}{\cos A - \sin A} + \frac{\sin A}{1 - \cot A} = \sin A + \cos A$$

iii) The triangle A (1,2), B(4,4) and C(3,7) is first reflected in line y =0 onto triangle A'B'C' and then triangle A'B'C' is reflected in the origin onto triangle A''B''C''. Write down the coordinate of [4]

a) A'B'C'

b) A''B''C''

c) Assign a special name to the quadrilateral BCC''B'' and find its area.

Question 5

i) Shown below is a table illustrating the monthly income distribution of a company for 100 employees [3]

Monthly Income (in ₹ 10,000)	0-4	4-8	8-12	12-16	16-20	20-24
Number of employees	55	15	06	08	12	4

Use step-deviation method to find the mean income of an employee.

ii) There are three positive numbers in a GP such that [3]
a) their product is 3375
b) the result of the product of first and second number added to the product of second and third number is 750.Find the numbers .

iii) Find the amount of bill for the following intra-state transaction of goods/services [4]

M.R.P(₹)	12000	15000	9500	18000
Discount %	30	20	30	40
CGST %	6	9	14	25

Question 6

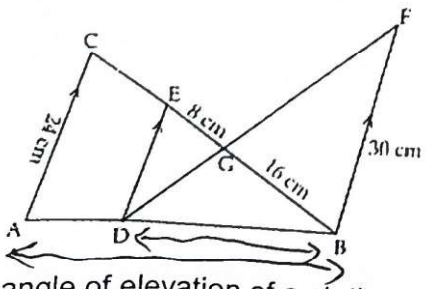
i) A man opened a recurring deposit account in a branch of PNB .The man deposits certain amount of money per month such that after 2 years ,the interest accumulated is equal to the monthly deposit .Find the rate of interest per annum that the bank was paying for recurring deposit account. [3]

ii) Solve for x if $\frac{5}{x} + 4\sqrt{3} = \frac{2\sqrt{3}}{x^2}$ [3]

iii) Use a ruler and a compass to answer this question .Construct a triangle ABC where AB= 5.5cm, BC = 4.5cm ,and $\angle ABC = 135^0$. [4]
Construct the circumcircle to the ΔABC . Measure and write down the length of AC.

Question 7

i) In the given figure, AC || DE || BF. If AC = 24 cm, EG = 8 cm, GB = 16 cm, BF = 30 cm [3]
(a) Prove ΔGED ~ ΔGBF
(b) Find DE
(c) DB : AB



) The angle of elevation of a stationary cloud from a point 25m above a lake is 30^0 and the angle of depression of its reflection in the lake is 60^0.What is the height of the cloud above the lake –level? [3]

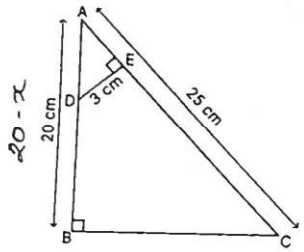
iii) Using ruler and compass only for answering this question .Mark a point P outside a circle of radius 5cm, at a distance of 7cm from the centre .Construct two tangents to the circle from the external point P. Measure and write down the length of only one tangent. [4]

Question 8

i) Draw an inscribing circle of a regular hexagon of side 7cm.Measure its radius. [3]

ii) The roots of the equation [3]
$(q-r)x^2+(r-p)x+p-q=0$ are equal .
Prove that p,q,r are in A.P.

iii) In ΔABC , $\angle ABC = 90°$, AB = 20 cm, AC = 25 cm. DE is perpendicular to AC such that $\angle DEA = 90°$ and DE = 3 cm as shown in the given figure [4]



a) Prove that $\Delta ABC \sim \Delta AED$.

b) Find the lengths of BC, AD and AE.

c) If BCED represents a plot of land on a map whose actual area on ground is 576 m^2, then find the scale factor of the map.

Question 9

i) a)If the lines $kx - y + 4 = 0$ and 2y = 6x +7 are perpendicular to each other, find the value of k . [3]

b) Find the equation of a line parallel to 2y =6x +7 and passing through (1,-1)

ii) Mr and Mrs Das was travelling by car from Delhi to Kasauli . Distance between Delhi and Kasauli is approximately 350 km. Due to heavy rain they had to slow down. The average speed of the car was reduced by 20km/h and the time of the journey increased by 2hrs. Find : [3]

a) original speed of the car

b) With the reduced speed, the number of hours they took to reach the destination.

iii) The mean of the following frequency distribution is 57.6 and the sum of all the frequencies is 50. Find the value of p and q. [4]

class	0-20	20-40	40-60	60-80	80-100	100-120
frequency	7	p	12	q	8	5

Question 10

i) Using remainder and factor theorem, show that (2x+3) is a factor of the polynomial $2x^2 + 11x + 12$. Hence factorize it completely .What must be multiplied to the given polynomial so that x^2+3x-4 is a factor of the resulting polynomial ? Also write the resulting polynomial. **[3]**

ii) The angle of elevation of the top of a tower from a point A on the ground is θ. The angle of elevation changes to $90 - \theta$ when the observer moves to a distance of x meters towards the base of the tower .Prove that the height of the tower is $\sqrt{d(d+x)}$ where d is final distance from the tower. **[3]**

iii) Vivek invests Rs.4500 in 8%, Rs.10 shares at Rs.15. He sells the shares when the price rises to Rs.30 and invests the proceeds in 12% Rs.100 shares at Rs.125. Calculate **[4]**

a) the sale proceeds

b) the number of Rs.125 shares he buys

c) the change in his annual income from dividends.

****

Test 8

Class: X 1st PRELIMINARY EXAMINATION

MATHEMATICS

Maximum Marks: 80

Time allowed: Three hours

Answers to this paper must be written on the paper provided separately. You will not be allowed to write during the first 15 minutes. This time is to be spent in reading the question paper. The time given at the head of this paper is the time allotted for writing the answers.

Attempt **all** questions from **Section A** and **any four** questions from **Section B.**

All working, including rough work, must be done on the same sheet as the rest of the answer.

Omission of essential working will result in loss of marks.

The intended marks for questions or parts of questions are given in brackets [ ].

Mathematical tables are provided.

SECTION A

(Attempt **all** questions from this Section)

Question 1

Choose the correct answers to the questions from the given options. [15]

(Do not copy the questions, write the correct answers only.)

(i) The marked price of an article is ₹ 5000. The shopkeeper gives a discount of 10%. If the rate of GST is 12%, then the amount paid by the customer including GST is:

(a) ₹ 5040

(b) ₹ 6100

(c) ₹ 6272

(d) ₹ 6160

(ii) Given, $x + 2 \leq \frac{x}{3} + 3$ and x is a prime number. The solution set for x is:

(a) $\emptyset$

(b) $\{0\}$

(c) $\{1\}$

(d) $\{0, 1\}$

(iii) Which of the following equation represents a line equally inclined to the axes?

(a) $2x - 3y + 7 = 0$

(b) $x = 7$

(c) $x - y = 7$

(d) $y = -7$

(iv) Mr. Lal deposits ₹ 3000 per month in a Recurring Deposit account for 2 years at 9% p.a., then the interest earned in 2 years will be:

(a) ₹ 6750

(b) ₹ 6570

(c) ₹ 6075

(d) ₹ 6705

(v) If A is 2×2 matrix whose elements are given by $a_{ij} = i - 2j$, then the value of a_{22} is:

(a) 2

(b) -2

(c) 1

(d) -1

(vi) For what value of x, $x - 3$, $2x + 1$ and $4x + 3$ are in AP?

(a) 0

(b) 1

(c) 2

(d) 3

(vii) 500, ₹ 50 shares at par earn a dividend of ₹ 1,250 in one year. The rate of dividend is:

(a) 10%

(b) 7·5%

(c) 12·5%

(d) 5%

(viii) If -2 is a root of the quadratic equation $x^2 + lx - 2 = 0$ and the quadratic equation $x^2 - lx + m = 0$ has equal roots, then the value of m is:

(a) 4

(b) $\frac{1}{4}$

(c) $\frac{-1}{4}$

(d) -4

(ix) If x, 4, 16 and y are in continued proportion then $2x - \frac{y}{32}$ is:

(a) 0

(b) 1

(c) $\frac{3}{2}$

(d) $\frac{2}{3}$

(x) Which of the following points is invariant with respect to the line $x = -2$?

(a) (3, 2)

(b) (3, −2)

(c) (2, 3)

(d) (−2, 3)

(xi) Out of 800 identical articles some are selected and the remaining are rejected. If the probability of an article to be rejected is 0.425; the number of selected articles is:

(a) 340

(b) 430

(c) 460

(d) 560

(xii) C_1 and C_2 are two cylindrical cans. A certain amount of water is filled in can C_1 up to the height 10 cm and then transferred in can C_2

Statement 1: The height of water in cylindrical can C_2 will be 2×10 cm = 20 cm.

Statement 2: The wetted surface area in both the cylinders will be same.

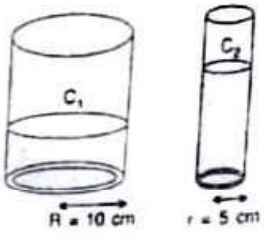
Which of the following is valid?

(a) Both statements are true

(b) Both statements are false

(c) Statement 1 is true and statement 2 is false

(d) Statement 1 is false and statement 2 is true.



(xiii) Points A and B have co-ordinates $(4, y)$ and $(x, 8)$. The mid-point of AB = (0, 0); the value of x and y are:

(a) $x = -4, y = 8$

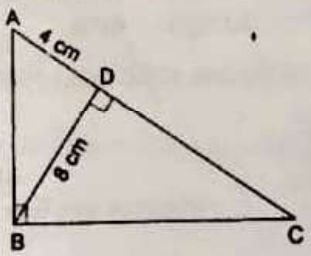
(b) $x = 4, y = -8$

(c) $x = -4, y = -8$

(d) $x = 4, y = 8$

(xiv) **Assertion (A) :** In the figure, $\angle ABC = \angle BDC = 90°$. If AD = 4 cm, BD = 8 cm, then area of ΔABC is 40 cm^2.

Reason (R) : Areas of two similar triangles are proportional to the squares of their corresponding sides.



(a) A is true, R is false

(b) A is false, R is true

(c) Both A and R are true

(d) Both A and R are false.

(xv) $\frac{1}{1-\sin A}$ is equal to:

(a) $\sec^2 A\,(1 + \sin A)$

(b) $(1 + \sin A)$

(c) $\sec^2 A$

(d) $\sec^2 A + \sin A$

Question 2

(i) Mr. Sen visits a mall and buys the following articles: [4]

Leather belt costing ₹ 950, GST at the rate of 5%.

A pair of shoes costing ₹ 3,000, GST at the rate of 18%

A laptop bag costing ₹ 1,000 with a discount of 30%, GST at the rate of 18%.

(a) Calculate the total amount of GST paid.

(b) The total bill amount including GST paid by Mr. Sen.

(ii) For what value of a, the polynomial $g(x) = 2x^2 - 3x$ is a factor of [4]

$f(x) = x^3 - 2x^2 + ax - 3a$?

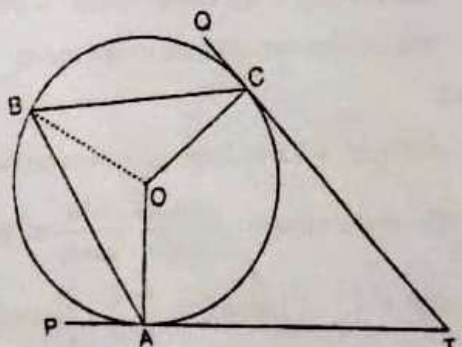
(iii) Determine the ratio in which the line $3x + y - 9 = 0$ divides the segment [4]

Joining the points (1, 3) and (2, 7).

Question 3

(i) In the given figure, TP and TQ are two tangents to the circle with centre O touching at A and C respectively. [4]

If $\angle BCQ = 55°$ and $\angle BAP = 60°$, find:



(a) $\angle OBA$ and $\angle OBC$

(b) $\angle AOC$

(c) $\angle ATC$.

(ii) A solid is composed of a cylinder with hemispherical ends. If the whole length of the solid is 105 cm and the diameter of the hemispherical ends is 36 cm each, find the cost (in nearest rupee) of polishing the surface of the solid at the rate of 21 paise per square cm. [4]

(iii) Use a graph paper for this question. [5]

The point P (5, 3) was reflected in the origin to get the image P'.

(a) Write down the co-ordinates of P'.

(b) If M is the foot of the perpendicular from P to the x -axis, find the co-ordinates of M.

(c) If N is the foot of the perpendicular from P' to the x -axis, find the co-ordinates of N.

(d) Name the figure PMP'N.

(e) Find the area of the figure PMP'N.

SECTION B

(Attempt **any four** questions from this Section)

Question 4

(i) A box contains 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, find the probability that it bears: [3]

(a) A two-digit number.

(b) A perfect square number.

(c) A number divisible by 5.

(ii) Solve the given inequation and graph the solution on the number line. [3]

$-2\frac{2}{3} \leq y + \frac{2}{3} < 3\frac{1}{3}, y \in Z.$

(iii) Mr. Negi sold a certain number of shares of ₹ 20 paying 8% dividend at ₹ 18 and invested the proceeds in ₹ 10 shares, paying 12% dividend at 50% premium. If the change in his annual income is ₹ 120, find the number of shares sold by Mr. Negi. [4]

Question 5

(i) Ashima has a Cumulative Deposit account in a bank for 5 years at 9% p.a. At the time of maturity, she gets ₹ 51,607·50. Find the monthly instalment. [3]

(ii) Find the value of p, if the lines, $5x - 3y + 2 = 0$ and $6x - py + 7 = 0$ are perpendicular to each other. Hence find the equation of a line passing through $(-2, -1)$ and parallel to $6x - py + 7 = 0$. [3]

(iii) The 4th, 6th and the last term of a geometric progression are 10, 40 and 640 respectively. If the common ratio is positive, find the first term, common ratio and the number of terms of the series. [4]

Question 6

(i) Solve $x^2 + 7x = 7$ and give your answer correct to two decimal places. [3]

(ii) Prove the identity: $\frac{\sin A - 2\sin^3 A}{2\cos^3 A - \cos A} = \tan A.$ [3]

(iii) If $A = \begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 1 \\ -3 & 2 \end{bmatrix}$ and $A^2 - 5B^2 = 5C$. Find matrix C where C is a 2×2 matrix. [4]

Question 7

(i) An observer on the top of a cliff; 200 m above the sea level, observes the angle of depression of the two ships to be 45° and 50° respectively. Find the distance between the ships (to the nearest metres), if the ships are on the opposite sides of the cliff. [4]

(ii) Use a graph sheet for this question. The daily wages of 120 workers working at a site are given below: [6]

Wages (₹)	250-300	300-350	350-400	400-450	450-500	500-550	550-600
Number of workers	8	15	20	30	25	15	7

Use 2 cm = ₹ 50 along one axis and 2 cm = 20 workers along the other axis to draw an ogive and hence estimate:

(a) The median wages.

(b) The inter quartile range of wages.

(c) Percentage of workers whose daily wage is above ₹ 475.

Question 8

(i) A person on tour has ₹ 360 for his expenses. If he extends his tour for 4 days, he has to cut down his daily expenses by ₹ 3. Taking the original duration of tour as x, form an equation in x and solve it. [3]

(ii) Find the sum: $1 + 7 + 13 + \ldots + 115$. [3]

(iii) If $x = \frac{\sqrt[3]{m+1} + \sqrt[3]{m-1}}{\sqrt[3]{m+1} - \sqrt[3]{m-1}}$, show that $x^3 - 3mx^2 + 3x - m = 0$. [4]

Question 9

(i) If the mean of the following frequency distribution is 7·5, find the value of p. [3]

x	3	5	7	9	11	13
f	6	8	15	p	8	4

(ii) If $(x - 2)$ is a factor of $2x^3 - x^2 - px - 2$, [3]

(a) Find the value of p.

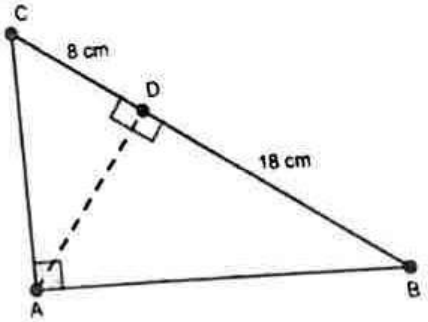
(b) Factorize the above expression completely.

(iii) In the adjoining figure, ABC is a right-angled triangle with $\angle BAC = 90°$ [4]

(a) Prove that $\Delta ADB \sim \Delta CDA$.

(b) If BD = 18 cm and CD = 8 cm, find AD.

(c) Find the ratio of the area of ΔADB and the area of ΔCDA.



Question 10

(i) Prove the identity: $\frac{\tan^2 A}{(\sec A - 1)^2} = \frac{1+\cos A}{1-\cos A}$ [3]

(ii) Use ruler and compass for this question. Construct a circle of radius 4·5 cm. [3]

Draw a chord AB = 6 cm.

(a) Find the locus of points equidistant from A and B. Mark the point where it meets the circle as D.

(b) Join AD and find the locus of points which are equidistant from AD and AB. Mark the point where it meets the circle as C. Join BC and CD.

(iii) In the given figure, ABCDE is a pentagon inscribed in a circle such that AC is a diameter and side BC ∥ AE. [4]

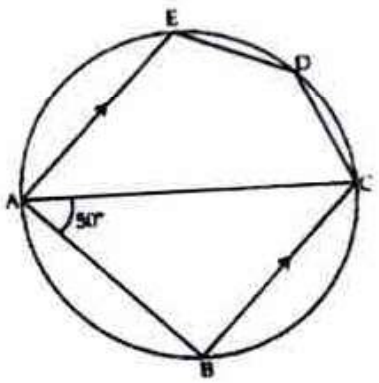
If ∠BAC = 50°, find:

(a) ∠ACB

(b) ∠EDC

(c) ∠BEC

Hence prove that BE is also a diameter.



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FIRST PRELIMINARY EXAMINATION

Paper: Mathematics

Grade: 10

Marks: 80

Time: 3 hours

Answer to this paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the question paper.
The time given at the head of this paper is the time allowed for writing the answers.

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*Attempt **all** questions from Section A and any **four** questions from Section B.*
All working, including rough work, must be clearly shown and must be done on the same sheet as the rest of the answer.
Omission of essential working will result in loss of marks.
The intended marks for questions or parts of a question are given in the brackets [ ]

SECTION – A (40 Marks)

*(Attempt **all** questions from this Section.)*

Question 1

Choose the correct answers to the questions from the given options: [15]

i) If $f(x)$ is a polynomial and a is a real number, then $x + a$ is the factor of $f(x)$ only when:
a) $f(a) = 0$ b) $f(-a) = 0$ c) $f(-a) = -a$ d) none of these

ii) Dia deposited ₹ 1000 per month in a recurring deposit account for 3 years, the rate of interest being 10% p.a.
(1) The total interest earned by Dia is ₹ 5550.
(2) Amount received on maturity is ₹ 36000.
Which of the following statement/s is/are valid?
a) Only 1
b) Only 2
c) Both 1 and 2
d) Neither 1 nor 2

iii) Two objects that have the same shape, but are different in size are said to be
a) Similar b) congruent c) a-like d) identical

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iv) Under GST 'value addition' refers to:

a) expenses plus profit b) cost plus tax plus profit

c) cost plus tax d) tax plus profit

v) The common ratio of a G.P. is 2 and its 6th term is 48. The first term is :

a) 3/2 b) 2/3 c) 1 d) 2

vi) The slope of the line $ax + by + c = 0$ is:

a) $\frac{a}{b}$ b) $-\frac{a}{b}$ c) $-\frac{c}{b}$ d) $\frac{c}{b}$

vii) Which statement is not correct?

a) Two matrices are added only when they are of same order.

b) A matrix is said to be zero matrix if all the elements are zero

c) A matrix is said to be a square matrix if the number of rows are equal to number of columns

d) A matrix of order $m \times n$ has $m + n$ elements

viii) The solution set of : $3x - 4 > x + 6; x \in N$ is :

(1) $x > 5$

(2) $x < 5$

Which of the following statement/s is/are valid?

a) Only 1 b) Only 2 c) Both 1 and 2 d) Neither 1 nor 2

ix) The sum of two whole numbers is 18 and their product is 45, the numbers are :

a) 15 and 3 . b) -15 and – 3 c) -15 and 3 d) 15 and - 3

x) In context of shares:

(1) If market value of a share is equal to face value of the share, the share is called at par.

(2) If market value of a share is less than the face value of the share, the share is called at discount or below par.

Which of the following statement/s is/are valid?

a) Only 1 b) Only 2 c) Both 1 and 2 d) Neither 1 nor 2

xi) The vertices of a triangle are (-5, 3), (p, -1) and (6, q) and the centroid is (1, -1). Find p and q.

a) 2 and -5 b) 2 and 5 c) -2 and 5 d) -2 and -5

xii) In an A.P., $a = -36$, $d = 18$ and $l = 36$, then n is:

a) 10 b) 15 c) 5 d) 20

xiii) If $(m + n) : (n - m) = 5 : 2$; m : n is :

a) 3 : 7 b) 7 : 3 c) 5 : 3 d) 3 : 5

xiv) Assertion (A) : $(2x - 1)^2 - 4x^2 + 5 = 0$ is not a quadratic equation.
Reason (R) : An equation of the form $ax^2 + bx + c = 0$, $a \neq 0$, where a, b, c $\in$ R is called a quadratic equation

a) A is true, R is false
b) A is false, R is true
c) Both A and R are true
d) Both A and R are false

xv) The coordinates of the point (-3, 4) after reflection in y-axis is :

a) (3, 4) b) (-3, 4) c) (3, - 4) d) (-3, -4)

Question 2

i) Mr. Mohan has a recurring deposit account in a post office for 3 years at 7.5% p.a. simple interest. If he gets ₹ 8,325 as interest at the time of maturity, find : [4]

a) the monthly instalment
b) the amount of maturity

ii) Prove that: [4]

$\cos A (1 + \cot A) + \sin A (1 + \tan A) = \sec A + \text{cosec } A$

iii) Calculate the ratio in which the line joining A (-4, 2) and B (3, 6) is divided by point P (x, 3). Also find: [4]

a) x
b) Length of AP

Question 3

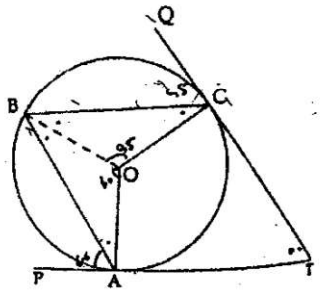
i) Solve the following inequation, write the solution set and represent it on the real number line: [4]

$$4x - 19 < \frac{3x}{5} - 2 \leq \frac{-2}{5} + x, \; x \in \mathbf{R}$$

ii) In the adjoining figure, TP and TQ are two tangents to the circle with centre O, touching at A and C respectively. [4]

If $\angle BCQ = 55°$ and $\angle BAP = 60°$, find :

(a) $\angle OBA$ and $\angle OBC$
(b) $\angle AOC$
(c) $\angle ATC$



iii) 100 pupils in a school have heights as tabulated below: [5]

Height in cm.	121-130	131-140	141-150	151-160	161-170	171-180
No. of pupils	12	16	30	20	14	8

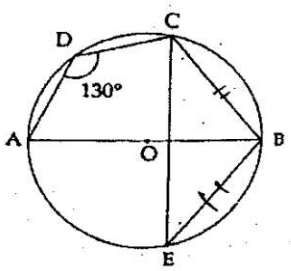
Draw the ogive for the above data.

SECTION B

(Attempt any 4 questions from this section.)

Question 4

i) In the adjoining figure, AB is a diameter of the circle with centre O. If $\angle ADC = 130°$ and chord BC = chord BE, find $\angle CBE$ [3]



ii) Write down the equation of the line whose gradient is 3/2 and which passes through P, where P divides the line segment joining A (-2, 6) and B (3, -4) in the ratio 2 : 3. [3]

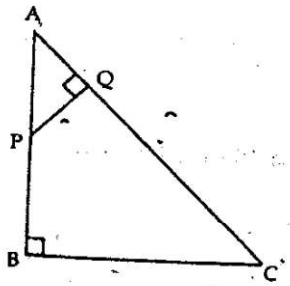
iii) How many terms of the A.P. 9, 17, 25, ... must be taken to give a sum of 636? [4]

Question 5

i) Find the number of coins, 2.4 cm in diameter and 2 mm thick, to be melted to form a right circular cylinder of height 12 cm and diameter 6 cm. [3]

ii) Given a G.P. with $a = 729$ and 7th term = 64, determine S_7 [3]

iii) In Δ ABC, AB = 8 cm, AC = 10 cm and $\angle B = 90°$. P and Q are points on the sides AB and AC respectively such that PQ = 2 cm and $\angle PQA = 90°$, find the area of Δ AQP. [4]



Question 6

i) When divided by $(x - 3)$, the polynomials $x^3 - px^2 + x + 6$ and $2x^3 - x^2 - (p + 3)x - 6$ leave the same remainder. Find the value of 'p'. [3]

ii) A man owns 560 shares of a company. The face value of each share is ₹ 25 and the company declares a dividend of 9% . Calculate : [3]

(a) the dividend the man would receive

(b) the rate of interest on his investment considering that the man bought these shares at ₹ 30 per share in the market.

iii) If $f(x) = x^2 - 5x + 7I$ and $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, find $f(A)$, where I is the identity matrix of order 2. [4]

[4]

Question 7

i) If $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, show that $\frac{x^3}{a^3} + \frac{y^3}{b^3} + \frac{z^3}{c^3} = \frac{3xyz}{abc}$ [5]

ii) Use graph paper for this question: [5]

a) The point P (2, -4) is reflected about the line $x = 0$ to get the image Q. Find the coordinates of Q.

b) Point Q is reflected about the line $y = 0$ to get the image R. Find the coordinates of R.

c) Name the figure PQR.

d) Find the area of figure PQR.

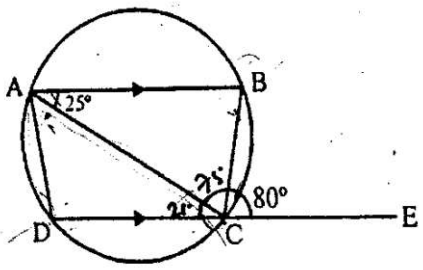
Question 8

i) Given that $\tan\theta = \frac{p}{q}$, find the value of $\frac{p\sin\theta - q\cos\theta}{p\sin\theta + q\cos\theta}$ [3]

ii) A toy is in the form of a cone mounted on a hemisphere of radius 3.5 cm. The total height of the toy is 15.5 cm. Find the total surface area. [3]

iii) In the given figure, AB is parallel to DC, $\angle BCE = 80°$ and $\angle BAC = 25°$. [4]

Find: a) $\angle CAD$ b) $\angle CBD$ c) $\angle ADC$



Question 9

i) Subodh purchases an article for ₹ 5400 which includes 10% discount on the marked price and 5% GST on the remaining price. Find the market price of the article. [3]

ii) Solve the following equation: $x - \frac{18}{x} = 6$. Give your answer correct to two decimal places. [3]

iii) The age of a man is twice the square of the age of his son. Eight years hence, the age of the man will be 4 years more than three times the age of his son. Find their present ages. [4]

Question 10

i) A die is rolled once. Find the probability of getting: [3]

a) a perfect square

b) an even prime number

c) a number < 5

ii) Construct a triangle ABC in which AB = 5 cm, BC = 8 cm and CA = 7 cm. Construct the circle which passes through A and B and which has BC as a tangent. [3]

iii) If $\frac{a^3+3ab^2}{3a^2b+b^3} = \frac{x^3+3xy^2}{3x^2y+y^3}$, using properties of proportion show that : $\frac{x}{a} = \frac{y}{b}$ [4]

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61.

[6]

Test 10

FIRST PRELIMINARY EXAMINATION
MATHEMATICS

STD: X

Maximum Marks: 80

Time allowed: Three hours

Answers to this paper must be written on the paper provided separately.

*You will not be allowed to write during the first **15 minutes.***

This time is to be spent in reading the question paper.

The time given at the head of this paper is the time allowed for writing the answers

---

*Attempt **all** questions from **Section A** and any four from **Section B.***

All working, including rough work, must be clearly shown and must be done on the same sheet as the rest of the answer.

Omission of essential working will result in loss of marks.

The intended marks for questions or parts of questions are given in brackets [ ].

*This question paper consists of **six** printed pages.*

Nothing should be written on the question paper.

Mathematical tables are provided.

Continued on page 2

Section A (40 Marks)

*Attempt **all** questions from this section.*

Question 1

Choose the correct answers to the questions from the given options.
(Do not copy the question, write the correct answers only.) **[15]**

(i) The probability of getting a number divisible by 3 in throwing a die is :

(a) $\frac{1}{6}$
(b) $\frac{1}{3}$
(c) $\frac{1}{2}$
(d) $\frac{2}{3}$

(ii) The volume of a conical tent is 462 m^3 and area of the base is 154 m^2. The height of tent is :

(a) 15 m
(b) 12 m
(c) 9 m
(d) 24 m

(iii) The median class for the given distribution is :

C.I.	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Frequency	2	4	3	5	1

(a) 0 – 10
(b) 10 – 20
(c) 20 – 30
(d) 30 – 40

(iv) If two lines are perpendicular to one another then the relation between m_1 and m_2 is :

(a) $m_1 = m_2$
(b) $m_1 = -m_2$
(c) $m_1 \times m_2 = -1$
(d) $m_1 \times m_2 = 1$

(v) A light house is 80 m high. The angle of elevation of its top from a point 80m away from its foot along the same horizontal line is :

(a) 60^0
(b) 45^0
(c) 30^0
(d) 90^0

(vi) The model class of a given distribution always corresponds to the :

(a) Interval with highest frequency
(b) Interval with lowest frequency
(c) The first interval
(d) The last interval

Continued on page 3

(vii) The co-ordinates of the point P(–3, 5) on reflecting x – axis are :

(a) (3, 5)

(b) (–3, –5)

(c) (3, –5)

(d) (–3, 5)

(viii) In cyclic quadrilateral ABCD, $\angle BAD = (2x + 5)^0$ and $\angle BCD = (x + 10)^0$ then the value of x is:

(a) 65^0

(b) 45^0

(c) 55^0

(d) 5^0

(ix) A (1,4), B (4,1) and C (x,4) are the vertices of ΔABC. If the centroid of the triangle is G(4,3) then the value of x is:

(a) 2

(b) 1

(c) 7

(d) 4

(x) The radius of a roller 100 cm long is 14 cm. The curved surface area of roller is :

(a) 13200 cm^2

(b) 15400 cm^2

(c) 4400 cm^2

(d) 8800 cm^2

(xi) Sarvesh sells a pen and paid CGST of ₹ 40 on which rate of GST is 16% then the profit of Sarvesh is:

(a) ₹ 800

(b) ₹ 500

(c) ₹ 600

(d) ₹ 300

(xii) **Assertion:** The equation $(3x+1)^2 - 9x^2 + 5 = 0$ is a quadratic equation.

Reason: Any equation of the form $ax^2 + bx + c = 0$,$a \neq 0$ is a quadratic equation.

(a) Both assertion and reason are correct and reason is the correct explanation of assertion.

(b) Both assertion and reason are correct but reason is not the correct explanation of assertion.

(c) Assertion is correct but reason is incorrect

(d) Assertion is incorrect but reason is correct

(xiii) The twelfth term from end of A.P. –2, –4, –6, ...–100 will be:

(a) –80

(b) –76

(c) –74

(d) –78

Continued on page 4

(xiv) If $\Delta ABC \sim \Delta DBA$, then:

(a) $AB^2 = AC \times BC$

(b) $AB^2 = BC \times DB$

(c) $AB^2 = AD \times DB$

(d) $AB^2 = AD \times AC$

(xv) 8th term of G.P. 5,10,20,40….. is :

(a) 1280

(b) 160

(c) 640

(d) 320

Question 2

(a) Let $A = \begin{bmatrix} 3 & 2 \\ 0 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$; Verify that $(A + B)(A - B) = A^2 - B^2$ **[4]**

(b) Prove that: $\frac{\sin A}{1+\cos A} + \frac{1+\cos A}{\sin A} = 2 \operatorname{cosec} A$ **[4]**

(c) If the 3rd term of a G.P. is 4, find the product of its first five terms. **[4]**

Question 3

(a) A (2a+2, 3), B (7, 4) and C (2a + 5, 2) are are collinear.
Find the value of 'a' and distance AC. **[4]**

(b) Calculate the mean of the following by using short-cut method. **[4]**

C.I.	5 – 14	15 – 24	25 – 34	35 – 44	45 – 54
f	2	6	4	8	4

(c) Use a graph paper for this question. **[5]**
Use scale : 2 cm = 1 unit along both x and y-axes.

(i) Plot the points O (0,0), A (–4, – 4), B (–4, 0) and C (0, – 3).

(ii) Reflect points A and B in the y-axis, name them as A' and B' and write their co-ordinates.

(iii) Give geometrical name of BACA'B'.

(iv) Find the area of geometrical figure.

Section B (40 Marks)

*Attempt any **four** questions from this section*

Question 4 10/10

(a) Anil gets ₹ 6,455 at the end of one year at the rate of 14% per annum in a Recurring Deposit Account. Find the monthly instalment. **[3]**

(b) Solve the following : $2x - \frac{1}{x} = 7$. Give your answer to three significant figures. **[3]**

(c) $\Delta ABC \sim \Delta DEF$, AB = (x – 0.5) cm, AC = 1.5x cm, DE = 9 cm and FD = 3x cm, find the lengths of AB , DF and A(ΔABC) : A(ΔDEF) **[4]**

Continued on page 5

Question 5 7|10

(a) Draw a histogram for the given data and find its mode : [3]

Wt. in kg.	40 - 44	45 – 49	50 - 54	55 – 59	60 – 64	65 – 69
No. of plants	10	30	26	22	12	4

(b) If $x = \frac{\sqrt{3a+2b} + \sqrt{3a-2b}}{\sqrt{3a+2b} - \sqrt{3a-2b}}$, prove that : $bx^2 - 3ax + b = 0$, where $a, b \neq 0$. [3]

(c) The angle of elevation of the top of a 100m high tree from two points A and B on the opposite sides of the tree are 52^0 and 45^0 respectively. Find the distance AB to the nearest metre. [4]

Question 6 10|10

(a) Solve : $2x^2 - 10x + 5 = 0$ and correct the value in two places of decimal [3]

(b) The die is thrown once. What is the probability of getting: [3]
(i) multiples of 2.
(ii) an even number greater than 4.
(iii) number not less than 2

(c) Solve the following in-equation and represent the solution set on the number line. [4]

$$\frac{7x}{3} < 2x + 3 \leq 3x, \; x \in R$$

Question 7

(a) A motor boat, whose speed is 9 km / hr in still water, goes 12 km down stream ad come back in total time of 3 hours. Find the speed of stream. [3]

(b) Find the sum of numbers, each divisible by 7, lie between 300 and 700 ? [3]

(c) Use factor theorem to factorise the following polynomial completely: [4]
$6x^3 + 17x^2 + 4x - 12$

Question 8

(a) Shan from Mumbai buys some items worth ₹ 5000 from Nagpur. If rate of GST is 40%.
Find (i) IGST (ii) CGST (iii) Amount of bill [3]

(b) P and Q are the centres of circles with radii 9 cm and 2 cm respectively. PQ = 17 cm.
R is the centre of circle with radius x cm, which touches the above circles externally.
Given that $\angle PRQ = 90^0$, find the radius of circle with centre R. [3]

(c) Oil stored in a spherical vessel occupying $\frac{3}{4}$ of its full capacity. [4]
Radius of this spherical vessel is 28 cm. Oil poured into a cylindrical vessel with radius 21 cm.
Find the height of the oil in the cylindrical vessel (correct to nearest cm).

Continued on page 6

Question 9

(a) The line joining the points A (4, 4) and B (–8, –8) is divided by the x-axis at point C. Find : (i) AC : BC (ii) Co-ordinates of C [3]

(b) ABC is an isosceles triangle with AB = AC. A circle through B touches side AC at its middle point D and intersects side AB in point P. Show that : $AB = 4 \times AP$ [3]

(c) Given two straight lines $3x - 2y = 5$ and $2x + ky + 7 = 0$. Find the value of k for which the given lines are perpendicular to each other and find the point of intersection of these lines. [4]

Question 10 7/10

(a) Aahana deposits ₹ 1000 every month in a recurring deposit account for 2 years. If she receives ₹ 26000 on maturity, find the rate of interest. [3]

(b) What number should be subtracted from each of the numbers 23, 30, 57 and 78 so that the remainders are in proportion? Write the numbers which are in proportion. [3]

(c) In the given figure, O is the centre of the circle and AB is the diameter. DO // CB and $\angle DCB = 120°$. [4]

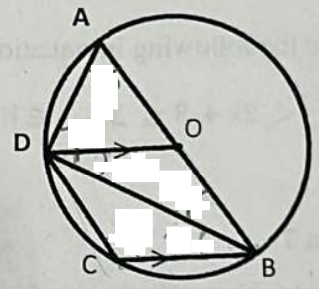
Calculate:

(a) $\angle DAB$

(b) $\angle DBC$

(c) $\angle ADC$

(d) $\angle DOB$



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